1 Transfer and Upload of Quality Management Data

1.1 Overview

This document describes how data is exchanged on a logical level between HYDRA Quality Management applications and an external system and which system components are required to do so.

1.2 Technical system requirements for communication

Data transfer and data processing is designed so that any external system can connect to the HYDRA server that can perform file transfer. Here, the physical connection between the host computer and the HYDRA server is not relevant.

To realize an automatic connection between the HYDRA sever and the host computer, MPDV recommends to use a gateway with the relevant control software. This gateway controls the data exchange between the two systems in both directions. But of course you can also use any other method that guarantees a secure file transfer, especially the connection of both systems using a "network file system".

1.3 Logical connection between the systems

All data is exchanged via file transfer using the physical system interface between the two systems. It then depends on the required response times and the physical connection between both systems how often the file transfer is performed.

All HYDRA postings to the host and vice versa are stored in transfer files in UTF-8 format (without BOM) and then transferred via file transfer.

This document describes the following transfer data:

Data class	Transfer direction	Input file	Output file
Article catalog	External system → HYDRA	as per parameters	-
Company catalog	External system → HYDRA	as per parameters	-
Inspection station catalog	External system → HYDRA	as per parameters	-
Failure analysis catalog	External system → HYDRA	as per parameters	-
Measures catalog	External system → HYDRA	as per parameters	-

Analysis selection catalog	External system → HYDRA	as per parameters	-
Entries from analysis selection catalogs	External system → HYDRA	as per parameters	-
Inspection plans	External system → HYDRA	as per parameters	-
Inspection plan characteristics	External system → HYDRA	as per parameters	-
Inspection plan/ inspection plan characteristics documents	External system → HYDRA	as per parameters	-
Specification list	External system → HYDRA	as per parameters	-
Inspection requirements	External system → HYDRA	as per parameters	-
Complaint header data	External system → HYDRA	as per parameters	-
Detailed complaints data	External system → HYDRA	as per parameters	-
Uploading inspection requirement results	HYDRA → external system	-	`pavrueck.asc`

The postings or records are stored on the individual systems in different files ("input and output file") so that they can be processed by separate programs. On the host computer, the files are either processed as a batch or with the help of an online transaction. The file processing itself is performed in the systems. For the data transferred to HYDRA, the system creates log and error files.

2 Data transfer from the external system

2.1 Description of the interface

This interface is realized using a new technology for universal interfaces. Using this technology, a separate command (so-called "dialogs") is listed in the interface file for each line to import the required data. The interface program processes these dialogs sequentially and writes a log file of the results.

Each line of the interface file contains a dialog including all relevant data. The most important components are the **dialog ID** and the **parameters** belonging to the dialog.

Example of a dialog (one line of an interface file):

```
DLG=CPAN.DELETE|CPAN.RECTYP=FEP|CPAN.BER=F|CPAN.PPS:REF=A449|...
```

Explanation:

The dialog ID is `CPAN.DELETE`. Separated by vertical lines ("|", ASCII 124), the parameters (`CPAN.BER`, `CPAN.PPS:REF`, ...) then follow. Select a field width for the different parameters that is wide enough for the presentation. However a fixed file structure can also be realized by adding leading zeros (for numbers) and trailing spaces.

The sequence of the parameters is irrelevant. Some parameters must be specified in a dialog and some parameters are optional. You can specify the optional parameters, if required, or you can leave these parameters out. HYDRA than populates these parameters using default values.

IMPORTANT! A dialog must end with a vertical line ("|", ASCII 124), otherwise what may happen is that the last parameter is ignored.

Please note with regard to fields with data types "N,<n>" and "N<x>.<y>".

If these fields are not included, they are filled with zeros by default. If the fields should be empty, the field must be transferred without content.

Example of a default assignment without content: `...|CMM.OTG=|...`

2.1.1 Conventions used to present the various data types

Data type	Format	Examples
N<n>	Numbers, a maximum of <n> positions	`|CPAN.PANNR=345678|` or `|CPAN.PANNR=   345678|` **Comment**: The preset value is "0" if fields of this type are not a part of the interface.
N<x>.<y>	Decimal number, a maximum of <x> positions before the comma. After the comma, only <y> positions are practical. The dot is the decimal separator.	`|CPAN.CMENGE=30.5|` or `|CPAN.CMENGE=00030.5|` **Comment**: The preset value is `"0.0"` if fields of this type are not a part of the interface.
C<n> Texts (characters)	Optional, the maximum length of <n> must be considered, though.	`|CPAN.PANVON=Huber|` or `|CPAN.PANVON=Huber   |`
Date	MM/DD/YYYY (American format)	`|CPAN.PANDAT=12/31/2001|`

Times or durati-
ons
	Seconds since 0:00
	or
	the following optional format, which is not supported with all time parameters:
	
	HH:MM or
	HH:MM:SS or
	HH,DDD or
	HH.DDD
	
	H hours (as many places as required)
	M Minutes (in groups of 60)
	S Seconds
	D Industrial or decimal minutes (in groups of 100)
		`|CPAN.PANZEI=52200|`
		or
		
		`|CPAN.PANZEI=14:30|` or
		`|CPAN.PANZEI=14:30:00|` or
		`|CPAN.PANZEI=014,5|` or
		`|CPAN.PANZEI=14.500|`

" "	Constant value	`|CARTIKEL.SMARTDEL=1|`

2.1.2 Conventions used to present mandatory fields

The acronyms in column "Mandatory" specify the dialogs with mandatory fields:

M for MODIFY

I for INSERT

U For UPDATE

C for COPY

D for DELETE

K for IFCMARK (inspection plans)

S for CANCEL (inspection requirements)

A for ACTIVATE (inspection requirements) or COMPLETING (inspection points)

R for RELEASE (inspection points)

F stands for "function" and means that this field is a mandatory field with MODIFY and INSERT if this function is used. The values preassigned to these fields can be different when data is created via interface or manually.

2.1.3 Calling the interface program

The interface program must be called from the HYDRA server. Processing is possible using the HYDRA scheduler.

Any existing data may be transferred from the external system into HYDRA using the program and call described below. As a general rule, this program must be called from the HYDRA server directory.

Windows systems as of HYDRA MW 2.0:

```
hymw.exe –u <UserNumber> -b <file name.ext>
```

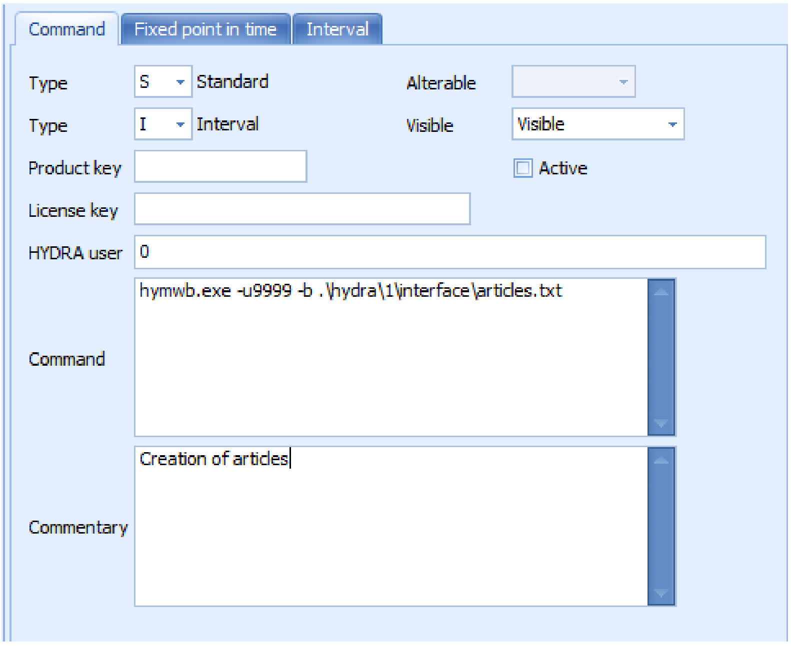
UNIX/AIX systems as of HYDRA MW 2.0:

```
hymw.out –u <UserNumber> -b <file name.ext>
```

If you add the parameter **-L,** the file is deleted after processing.

Here, a unique numeric value must be selected as the <UserNumber>. If interfaces with the same UserNumber run at the same time, they will interfere with each other and supply false results. Recommended are numbers ranging between 1 to 200.

Data is transferred from the specified file. Any errors that occur while processing this file will be stored in the error log **`<Filename>.err`** in the **`err`** subdirectory of the HYDRA directory. This file includes the record number and an error text relating to the erroneous record. The results of the processing can also be evaluated on the MOC via the HYDRA system logs under the abbreviation <Filename> (without extension). In addition, the system creates a log file (**`<FileName>.pro`**) of the action in the HYDRA subdirectory **`prot`**.



Example of a scheduler entry for the automatic generation of article data via interface

The entry ".\hydra\1\interface\articles.txt" is an example of how the path and the name of the interface file is specified.

The scheduler entry must not include a product or license key entry.

2.2 Description of the data structures

2.2.1 Data structures for multi-language / MDBI systems

If the HYDRA system is operated with multiple language database content in addition to the “native” default content, then each dialog line must contain the parameter

... | LANG=<x> | ...

for the respective target language. <x> stands for the number of the target language. For example, “German” language has the fixed language ID “1”, “English” is “2”, “Chinese” is “15”, etc.

If a data record/dialog shall be imported for the English database content in a multi-language system, then each dialog/data record must contain the parameter ...... | LANG=2 | ...... somewhere within the dialog string. We recommend putting it at the very right end of each dialog string.

This requirement applies for all described data structures/dialogs in multi-language systems, and is not explicitly mentioned in each sub-chapter!

2.2.2 Article catalog

The following dialogs are available for updating article master data:

- ARTIKEL.MODIFY to create or change article data
- ARTIKEL.INSERT to create article data
- ARTIKEL.UPDATE to change article data
- ARTIKEL.DELETE to delete article data

The unique key for the article catalog is made up of a combination of the fields ARTIKEL.ATK and ARTIKEL.ATKIDX.

If an article is already in use in HYDRA, it must not be deleted. For this reason, always pass the parameter ARTIKEL.SMARTDEL=1 when you delete data. This ensures that the archive flag (ARTIKEL.ARCHIV) is set if an article is used.

Please note for the group assignment or group generation

If you assign a group to an article, this group must have been created in HYDRA beforehand and the group name/designation must have been specified. If the assigned group includes several hierarchy levels, then a group with group name must exist for each combination of levels. The client shows the group designation/name assigned to each combination in the respective tree structure level.

Example: An article is assigned to a group with the fields "*.DSGRP.1", "*.DSGRP.2" and "*.DSGRP.3". In this case, a group must first be created solely for DSGRP1. In addition, there will also be the combinations "DSGRP1 + DSGRP2" and "DSGRP1 + DSGRP2 + DSGRP3".

Dialog: ARTIKEL.*				
Parameter	**Type**	**Man-dato-ry**	**Contents**	**Description**
*.ATK	C50	M/I/UD	Article number	
*.ATKIDX	C50	M/I/U/D	Drawing issue number	
*.DSGRPE	"0"	M/I	Fixed identification for article data	0 = Article, 1 = Article group

*.DSGRP:1	C50		Article group 1	Only required if article groups are used. In the dialog data string, you must pass the group ident number, which has been created previously, in this field. When an article group is created, HYDRA automatically assigns this ident number and for this reason it is recommended to assign the article groups via interface as well. You can then assign the article group ident number yourself.
*.DSGRP:2	C50		Article group 2	
*.DSGRP:3	C50		Article group 3	
*.DSGRP:4	C50		Article group 4	
*.DSGRP:5	C50		Article group 5	
*.ATKBEZ	C250		Article designation/name	
*.MOD	C250		Model	
*.ABC	C250		ABC article	
*.CEINH	C50		Unit of measure or size	A respective measurement unit with this identifier must already exist in HYDRA
*.ZEICHNR	C250		Drawing number	
*.DOKPFL	"0" or "1"	F	Documentation required	If the value "1" is transferred to this parameter, the article requires documentation.
*.ATK:KDNR	C50		Customer article number	
*.FU:1 to *.FU:6	Date		Direct user fields	These fields are only available if the license for "composition" is available or as of CAQ 8.2.
*.FU:7 to *.FU:22	N9		Direct user fields	These fields are only available if the license for "composition" is available or as of CAQ 8.2.
*.FU:23 to *.FU:28	N11.6		Direct user fields	These fields are only available if the license for "composition" is available or as of CAQ 8.2.
*.FU:29 to *.FU:44	C 1		Direct user fields	These fields are only available if the license for "composition" is available or as of CAQ 8.2.
*.FU:45 to *.FU:50	C10		Direct user fields	These fields are only available if the license for "composition" is available or as of CAQ 8.2.

*.FU:51 to *.FU:64	C 20		Direct user fields	These fields are only available if the license for "composition" is available or as of CAQ 8.2.
*.FU:65 to *.FU:66	C 40		Direct user fields	These fields are only available if the license for "composition" is available or as of CAQ 8.2.
*.ARCHIV	"0" or "1"	F	Archive flag	If the value "1" is transferred to this parameter, the relevant data record is marked as archived. This means this data record will not be available in selection lists.
*.SMARTDEL	„1“	D	Activate "smart" delete.	

2.2.3 Company catalogs

The following dialogs are available for updating company master data:

- FIRMA.MODIFY to create or change company data
- FIRMA.INSERT to create company data
- FIRMA.UPDATE to change company data
- FIRMA.DELETE to delete company data

The unique key for the company catalog is made up of a combination of the fields FIRMA.FIRTYP and FIRMA.FIRID.
If a company has already been used in HYDRA, it must not be deleted. This is why when deleting data, you should also always transfer the parameter FIRMA.SMARTDEL=1. Doing so will set the archive flag (FIRMA.ARCHIV) if an article is ever used.

Dialog: FIRMA.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**

*.FIRTYP	C50	M/I/U/D	Company type	"INTERNAL" "CUSTOMER", "SUPPLIER" or "MANUFACTURER"
*.FIRID	C50	M/I/U/D	Company number	
*.DSGRPE	"0"	M/I	Fixed identification for company data	
*.FIRBEZ	C250		Company name	
*.LAND	C250		Country	
*.STAAT	C250		Federal state, country or similar	
*.PLZ	C250		Zip code	
*.ORT	C250		City	
*.POSTFACH	C250		P.O. box	
*.ADRESSE:1	C250		Address	First line in the address area of forms (usually the company name)
*.ADRESSE:2	C250		Address	Second line in the address area of forms.
*.ADRESSE:3	C250		Address	Third line in the address area of forms.
*.TEL	C250		Phone number	
*.FAX	C250		Fax number	
*.HANDY	C250		Mobile phone number	
*.PAGER	C250		Pager number	
*.EMAIL	C250		e-mail address	
*.WWW	C250		Website address	

*.AUDIT:WERT	N12.4		Percentage of the last audit classification	
*.AUDIT:EINST	C50		Last audit classification	
*.AUDIT:DAT	Date		Date of the last audit classification	
*.ARCHIV	"0" or "1"	F	Archive flag	If the value "1" is transferred to this parameter, the relevant data record is marked as archived. This means this data record will not be available in selection lists.
*.SMARTDEL	„1“	D	Activate "smart" delete.	
*.VERANT:COPY	"0" or "1"	F	use as party in charge	If the value "1" is transferred to this parameter, the corresponding company will be assumed in the list of responsible parties.

2.2.4 Inspection station catalog

The following dialogs are available for updating inspection stations:

- `CPPLATZ.MODIFY` to create or change inspection station data
- `CPPLATZ.INSERT` to create or change inspection station data
- `CPPLATZ.UPDATE` to create or change inspection station data
- CPPLATZ.DELETE to delete inspection station data

The company catalog's unique key is the field `CPPLATZ.PPLATZ.` If an inspection station is already in use in HYDRA, it must not be deleted. This is why when deleting data, you should also always transfer the parameter `CPPLATZ.SMARTDEL=1.` Doing so will set the archive flag (`CPPLATZ.ARCHIV`) if an inspection station is ever used.

Dialog: `CPPLATZ.*`

Parameter	Type	Mandatory	Contents	Description
*.PPLATZ	C50	M/I/U/D	Inspection station number	
*.DSGRPE	"0"	M/I	Fixed identification for inspection station data	
*.PPLATZBEZ	C250		Inspection station designation/name	
*.ARCHIV	"0" or "1"	F	Archive flag	.
*.SMARTDEL	„1“	D	Activate "smart" delete.	

2.2.5 Failure analysis catalog

The following dialogs are available for updating the failure analysis catalog:

- CERRKAT.MODIFY to create or change failure analysis data
- CERRKAT.INSERT to create failure analysis data
- CERRKAT.UPDATE to change failure analysis data
- CERRKAT.DELETE to delete failure analysis data

The unique key for the failure analysis catalog is made up of a combination of the fields CERRKAT.ERRTYP and CERRKAT.ERRNR.
If a failure analysis criterion is already in use in HYDRA, it must not be deleted. This is why when deleting data, you should also always transfer the parameter CERRKAT.SMARTDEL=1. Doing so will set the archive flag (CERRKAT.ARCHIV) if an entry is ever used.

As concerns the group assignment or group creation, observe the notes in the chapter "Article catalog".

Dialog: `CERRKAT.*`				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
`*.ERRTYP`	C50	M/I/U/D	Analysis type	An analysis type with the corresponding abbreviation must be defined in HYDRA (status type "FHLANTYP"). The default types are listed below: "FA" Failure type "FO" Failure location "FU" Failure cause "VU" Causer/origin
`*.ERRNR`	C50	M/I/U/D	Entry number	
`*.DSGRPE`	"0"	M/I	Fixed identification for analysis data	0 = Failure, 1 = Failure group
`*.DSGRP:1`	C50		Analysis group 1	Only necessary if analysis groups are used.
`*.DSGRP:2`	C50		Analysis group 2	
`*.DSGRP:3`	C50		Analysis group 3	
`*.DSGRP:4`	C50		Analysis group 4	
`*.DSGRP:5`	C50		Analysis group 5	
`*.ERRBEZ`	C250		Name of analysis criterion	
`*.ARCHIV`	"0" or "1"	F	Archive flag	If the value "1" is transferred to this parameter, the relevant data record is marked as archived. This means this data record will not be available in selection lists.
`*.SMARTDEL`	„1“	D	Activate "smart" delete.	

2.2.6 Measures catalog

The following dialogs are available for updating the catalog of measures:

- `CMASKAT.MODIFY` to create or change measures in the catalog
- `CMASKAT.INSERT` to create measures in the catalog
- `CMASKAT.UPDATE` to change measures in the catalog
- `CMASKAT.DELETE` to delete data in the catalog of measures

The unique key for the measures catalog consists of the field `CMASKAT.MASNR`. If a catalog measure is already in use in HYDRA, it must not be deleted. This is why when deleting data, you should also always transfer the parameter `CMASKAT.SMARTDEL=1`. Doing so will set the archive flag (`CMASKAT.ARCHIV`) if an entry is ever used.

As concerns the group assignment or group creation, observe the notes in the chapter "Article catalog".

Dialog: CMASKAT.*				
Parameter	**Type**	**Man-dato-ry**	**Contents**	**Description**
*.MASNR	C50	M/I/U/D	Measure number	
*.DSGRPE	"0"	M/I	Fixed identification for catalog measures	0 = Measure, 1 = Measures group
*.DSGRP:1	C50		Measures group 1	Only necessary if measures groups are used.
*.DSGRP:2	C50		Measures group 2	
*.DSGRP:3	C50		Measures group 3	
*.DSGRP:4	C50		Measures group 4	
*.DSGRP:5	C50		Measures group 5	
*.MASBEZ	C250		Measures designation/name	

*.ARCHIV	"0" or "1"	F	Archive flag	If the value "1" is transferred to this parameter, the relevant data record is marked as archived. This means this data record will not be available in selection lists.
*.SMARTDEL	„1"	D	Activate "smart" delete.	

2.2.7 Analysis selection catalog

The following dialogs are available for updating the analysis selection catalog:

- CANAUSW.MODIFY to create or change analysis selection catalogs
- CANAUSW.INSERT to create analysis selection catalogs
- CANAUSW.UPDATE to change analysis selection catalogs
- CANAUSW.DELETE to delete analysis selection catalogs

The unique key for the analysis selection catalog consists of the field CANAUSW.ANAUSNR.
If an analysis selection catalog is already in use in HYDRA, it must not be deleted. This is why when deleting data, you should also always transfer the parameter CANAUSW.SMARTDEL=1. Doing so will set the archive flag (CANAUSW.ARCHIV) if an entry is ever used.

Dialog: CANAUSW.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
*.ANAUSNR	C50	M/I/U/D	Number of the analysis selection catalog	
*.ANAUSBEZ	C250		Name of the analysis selection catalog	
*.ARCHIV	"0" or "1"	F	Archive flag	If the value "1" is transferred to this parameter, the relevant data record is marked as archived. This means this data record will not be available in selection lists.

`*.SMARTDEL`	„1“	D	Activate "smart" delete.	

2.2.8 Entries from analysis selection catalogs

The following dialogs are available for updating entries in the analysis selection catalogs:

- `CANAWEINT.INSERT` to create entries for the analysis selection catalogs
- `CANAWEINT.DELETE` to delete entries for the analysis selection catalogs

The unique key for the entry of an analysis selection catalog consists of the fields `CANAWEINT.ANAUSNR`, `CANAWEINT.FLTTYP`, `CANAWEINT.EINTTYP` and `CANAWEINT.EINTNR`.

Dialog: `CANAWEINT.*`				
Parameter	**Type**	**Man-dato-ry**	**Contents**	**Description**
`*.ANAUSNR`	C50	I/D	Number of the analysis selection catalog	An analysis selection catalog with the respective number must exist in HYDRA.
`*.FLTTYP`	C50	I/D	Filter type	A filter type with the corresponding abbreviation must be defined in HYDRA (status type "ANAWFLTTYP"). These are the default types: "INCLUDE" inclusive "EXCLUDE" exclusive
`*.EINTTYP`	C50	I/D	Type of entry	An entry type with the corresponding abbreviation must be defined in HYDRA (status type "ANAWEINTTYP"). These are the default types: "FA" Failure type "FO" Failure location "FU" Failure cause "VU" Causer/origin "MASSKAT" Measure

`*.EINTNR`	C50	I/D	Entry identifier	A corresponding entry with the respective number must exist in HYDRA. The EINTTYP parameter specifies the origin of the entry.

2.2.9 Characteristics catalog

The following dialogs are available for updating catalog characteristics:

- `CMM.MODIFY` to create or modify catalog characteristics
- `CMM.INSERT` to create catalog characteristics
- `CMM.UPDATE` to change catalog characteristics
- `CMM.DELETE` to delete catalog characteristics

The unique key for a catalog characteristic consists only of the field `CMM.CMMNR`.

Dialog: `CMM.*`				
Parameter	**Type**	**Man-dato-ry**	**Contents**	**Description**
`*.CMMNR`	C50	M/I/U/D	Characteristic number	
`*.MMBEZ`	C250		Characteristic designation/name	
`*.MMTYP`	C50	F	Characteristic type	The parameter used here must be defined in HYDRA. Possible values: "PRODUKT" – Product characteristic "PROZESS" – Process characteristic
`*.ERFART`	"MANUELL"	F	Fixed identifier for CAQ characteristics	

`*.MUSSPRF`	"0" or "1"	F	Mandatory inspection	If this parameter has the value 1, an inspection step cannot be completed until at least one measured value was recorded for this characteristic.
`*.BFORMEL`	C250		Formula	If this parameter is completed with a value, then the characteristic's values will be calculated based on the formula defined here. In this case, it might no longer be possible to enter measured values for this characteristic manually (depends on the type of formula). Refer to the corresponding documentation to see the structure of a formula of this kind.
`*.PRUEFTYP`	C50	F	Characteristic type	The parameter used here must be defined in HYDRA. Possible values: "A" - attributive "V" - variable "F" – inspection chart (chart of recorded defects)
`*.ERFASSDET`	C50		Kind of input (inspection) type	This parameter is only available as of CAQ 8.2. The parameter used here must be defined in HYDRA. Possible values: "" – Standard "RASTER" - Visual defects recording → only supported with CMM.PRUEFTYP=F "CODE" – based on catalogs → only supported with CMM.PRUEFTYP=A "CODE_ZUFALL" – based on catalogs (random) → only supported with CMM.PRUEFTYP=A

*.BEWKAUSWMEN:1	C10		Selected set for assessment catalog	This parameter is only available as of CAQ 8.2. This parameter is only supported with CMM.ERFASSDET=CODE or CMM.ERFASSDET=CODE_ZUFALL.
*.PRFRASTER:X	C250		Grid of x-axis	This parameter is only available as of CAQ 8.2. This parameter is only supported with CMM.ERFASSDET=RASTER. The identifiers of the grid must be separated by comma.
*.PRFRASTER:Y	C250		Grid of y-axis	This parameter is only available as of CAQ 8.2. This parameter is only supported with CMM.ERFASSDET=RASTER. The identifiers of the grid must be separated by comma.
*.PMID	C50		Type PRM (test equipment/gage) resource to be used	If this parameter is specified, a respective PRM type resource must be defined in HYDRA. If this entry is used, the parameter CMM.RESFAM must be empty.
*.RESFAM			Resource family to be used	If this parameter is specified, a respective resource family must be defined in HYDRA. If this entry is used, the parameter CMM.PMID must be empty.
*.OPT:PLAN	M or G	F	M = Machine G = Machine group	You define here if the scheduling is performed for a machine/workplace or a group of machines/workplaces. M is the default value.
*.MNR	C50		Scheduling for a machine/workplace	The inspection step generated for this characteristic can only be logged on to the specified machine and inspected here.

*.MGRP	C 20		Scheduling for the specified machine/workplace group	The inspection step generated for this characteristic can be found in the sequencing list of all machines/workplaces that are included in this group. The inspection can therefore be performed with all machines of this group.
*.ZERTPRN	C50	F	Certificate printing	The ID entered here must be defined in HYDRA. Default values are: "NIE" – never print "AUSW" – selectable "IMMER" – always print
*.ERRGEW	C50	F	Failure weighting	The error weighting entered here must be defined in HYDRA. Default values are: "NEBEN" – Minor defect "HAUPT" –Major defect "KRIT" – Critical defect
*.BPRUEFERG	C50		Inspection result base	Specifies the basis for identifying the inspection result of the characteristic. NCD_ALL = inspection result is calculated over all samples NCD_LAST = inspection result is calculated from the last sample
*.ANAUSNR	C50		Analysis selection catalog	If this parameter is specified, a respective analysis selection catalog must be defined in HYDRA.
*.CEINH	C50		Unit of measure or size	A corresponding unit of measure with this identifier must already exist in HYDRA.

*.FMT	C250		Measured values format of the single values	Missing or completing zeros to be suppressed are marked by the "#" sign, positions that must definitely be displayed are illustrated with a "0". The decimal point is marked by a dot and the thousands separator is marked by a comma. Example: "#,##0.00##" This parameter is only needed for variable characteristics. If this parameter is not transferred, then the default format defined in the options is used. On the client, the number of decimal places is output as the format. This parameter is only required for variable characteristics.
*.GWNORM	C50		Standard	This parameter can be used to define which standard the tolerance limits are based on. A relevant standard must be defined in HYDRA. Standards are: • "ISO_PASS“ • "ISO7168“ • "ISO2768“ • "EN12420“ This parameter is only required for variable characteristics.
*.GWID	C50		Standard entry identifier	For some standards, an identifier can be specified for the corresponding entry. Example: "H7" for ISO fit standards This parameter is only required for variable characteristics.

`*.STPRPLAN`	C50	M/I (see comments)	Sampling scheme identifier	The sampling scheme must be defined in HYDRA. Possible values: • "NC" • "100PRO" • "SPC" • "LOS" The "LOS" sampling scheme is only valid in the goods receipt or the goods issue areas, while the sampling scheme "SPC" is only valid in the production areas.
`*.STPRUMF`	N9	M/I (see comments)	Sample size	This parameter determines the sample size. The value 0 must be used for open samples. The sampling scheme "100PRO" requires an entry (must not be empty)
`*.RWMENGE`	N9		Acceptance quantity	For STPRPLAN<>NC, the parameter must be empty
`*.RUMENGE`	N9		Rejection quantity	For STPRPLAN<>NC, the parameter must be empty
`*.INTTYP`	C50	F	Type of interval	With STPRPLAN<>SPC or STPRPLAN<>NC, the parameter must be empty. Valid values: • "KEINS" for no interval • "ZEIT" for time intervals • "STCK" for piece intervals • "EINMAL" for "once"
`*.INTERVAL`	N9		Interval	With STPRPLAN<>SPC or STPRPLAN<>NC, the parameter must be empty. Interval value. For piece intervals, the number of units is set in this field, for time intervals, the interval from this field results from the connection with the field CMM.INTEINH.

*.INTEINH	C50	F	Type of interval	With STPRPLAN<>SPC or STPRPLAN<>NC, the parameter must be empty. Specifies for time intervals which unit is found in the field CSPEZL.INTERVAL. Only identifiers defined in HYDRA can be used. By default, these are the following: • "SEK" – Seconds • "MIN" – Minutes • "STD" – Hours • "TAG" – Days • "MON" – Months • "JAH" – Years
*.INT:ALW	"0" or "1"	F	Characteristic becomes due for inspection when output batches are changed	Provided that MPL is in use
*.INT:MSW	"0" or "1"	F	Characteristic becomes due for inspection when machine statuses are changed	
*.INT:MSWMST	C250		Source status, target status or a combination of source and target sta-tus triggering inspec-tions when changing the machine status	Separate configurations by comma, no blanks
*.INT: SCHICHTW	"0" or "1"	F	Characteristic becomes due for inspection when the shift is changed	
*.PRBZUG	"0" or "1"	F	Sampling	This parameter can be used from SP4 onwards.
*.PRBGRP	C50		Sample group	This parameter can be used from SP4 onwards.

*.MASSANG	C50	F	Type of defined construction dimensions	Specifies the type how the construction dimensions are stored. Only identifiers defined in HYDRA can be used. By default, these are the following: • "ABSOLUT" – the construction dimensions are defined as absolute values in the parameters OPG, OTG, UTG and UPG. • "RELATIV" - the construction dimensions are defined as a relative deviation from the target value in the parameters OPGREL, OTGREL, UTGREL and UPGREL. • "PROZENTUAL" - the constructional measures are defined as a deviation in percent from the target value in the parameters OPGREL, OTGREL, UTGREL and UPGREL.
*.OPG	N12.4		absolute, upper plausibility limit	The parameter must be empty for PRUEFTYP<>V or MASSANG<>ABSOLUT .
*.OPGREL	N12.4		relative/ percentage, upper plausibility limit	The parameter must be empty for PRUEFTYP<>V or MASSANG=ABSOLUT .
*.OTG	N12.4		absolute, upper tolerance limit	The parameter must be empty for PRUEFTYP<>V or MASSANG<>ABSOLUT .
*.OTGREL	N12.4		relative/percentage, upper tolerance limit	The parameter must be empty for PRUEFTYP<>V or MASSANG=ABSOLUT .

*.OTGAKTIV	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V . If the value "1" is transferred to this parameter, a failure entry is automatically generated if single values violate the tolerance limit.
*.SW	N12.4		Target value	The parameter must be empty for PRUEFTYP<>V .
*.UTG	N12.4		absolute, lower tolerance limit	The parameter must be empty for PRUEFTYP<>V or MASSANG<>ABSOLUT .
*.UTGREL	N12.4		relative/ percentage, lower tolerance limit	The parameter must be empty for PRUEFTYP<>V or MASSANG=ABSOLUT .
*.UTGAKTIV	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V . If the value "1" is transferred to this parameter, a failure entry is automatically generated if single values violate the tolerance limit.
*.UPG	N12.4		absolute, lower plausibility limit	The parameter must be empty for PRUEFTYP<>V or MASSANG<>ABSOLUT .
*.UPGREL	N12.4		relative/ percentage, lower plausibility limit	The parameter must be empty for PRUEFTYP<>V or MASSANG=ABSOLUT .
*.KARTE:1	C50	F	Identifier for the first control chart	The following values are valid for variable characteristics: "XQ", "R", "S", "X" or "X_MED" The following values are valid for attributive characteristics: "P" or "U"
*.OEG:1	N12.4		upper action limit of the first control chart	

*.OEGAKTIV:1	"0" or "1"	F	Automatic failure entry	If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper action limit is violated by the statistical value of control chart 1 (e.g. Xq).
*.OWG:1	N12.4		upper warning limit of the first control chart	
*.OWGAKTIV:1	"0" or "1"	F	Automatic failure entry	If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper warning limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).
*.MWAVG:1	N12.4		Mean value for the first control chart	
*.UWG:1	N12.4		lower warning limit of the first control chart	The parameter must be empty for PRUEFTYP<>V .
*.UWGAKTIV:1	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V . If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower warning limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).
*.UEG:1	N12.4		lower action limit of the first control chart	The parameter must be empty for PRUEFTYP<>V .
*.UEGAKTIV:1	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V . If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower action limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).
*.VORG:1	C50	F	Default value for limit value calculation	The following values are valid: "CPK“, "SIGMA“, "QUER“, “ABW_REL“, "ABW_PROZ“

`*.CPK:1`	N12.4		CPK default value for limit value calculation	To calculate Sigma from the defined cpk, the upper and lower tolerance limits must be specified and must be distinguishable.
`*.STATVERT:1`	C50	F	Unilateral or bilateral calculation of limit values	The following values are valid: "EINS", "ZWEI". If this parameter has the value "EINS", the calculation is only performed for either the upper or the lower action/ warning limits. This setting is mandatory for attributive control charts. For R and s control charts, the upper limit values are calculated in this case. For Xq control charts, the calculation performed depends on which tolerance limit is specified. If the upper and lower tolerance limit is specified, the upper and lower warning/ action limits are calculated, even though "unilateral" was selected. If this parameter has the value "ZWEI", the calculation is performed for the upper and lower action/ warning limits.
`*.EWEG:1`	C50	F	Non-action probability for the action limits	The following values are valid for unilateral control charts: "1", "1.28", "1.64", "1.96", "2", "2.33", "2.58", "3", "3.09", "3.72", "4" The following values are valid for bilateral control charts: "1", "1.28", "1.64", "1.96", "2", "2.28", "2.33", "2.58", "3", "3.09", "3.45", "3.72", "3.9", "4" The following values are valid for R and s charts: "0.9", "0.95", "0.99"
`*.EWWG:1`	C50	F	Non-action probability for warning limits	See EWEG:1 parameters

*.RELEG:1	N12.4		relative deviation/ deviation in percent of the action limit(s) of the first (Xq) control chart from the target value	The parameter must be empty for KARTE:1<>XQ
*.RELWG:1	N12.4		relative deviation/ deviation in percent of the warning limit(s) of the first (Xq) control chart from the target value	The parameter must be empty for KARTE:1<>XQ
*.QUER:1	N12.4		Default value for sq: or rather for Rq: for computing limit values	
*.SIGMA:1	N12.4		Sigma default value for limit value calculation	
*.XQVART:1	C50	F	Type of Xq default value (for limit value calculation)	The following values are valid: "RKXQMITTE", "SOLLWERT", "TOLMITTE", "VORGABE"
*.VORGXQ:1	N12.4		Default value Xq for limit value calculation	For VORGXQ:1<>VORGABE, the parameter must be empty.
*.MOD:BER_WG_1	"0" or "1"	F	Automatically calculate warning limits for the first control chart	If the value "1" is transferred to this parameter, the warning limits for the first control chart are calculated automatically. It is not possible to calculate limit values for the Median chart.
*.MOD:BER_EG_1	"0" or "1"	F	Automatically calculate action limits for the first control chart	If the value "1" is transferred to this parameter, the action limits for the first control chart are calculated automatically. It is not possible to calculate limit values for the Median chart.
*.KARTE:2	C50	F	Identifier of the second control chart	The following values are valid for variable characteristics: "XQ", "R", "S", "X" or "X_MED" The following values are valid for attributive characteristics: "P" or "U"
*.OEG:2	N12.4		upper action limit of the second control chart	

*.OEGAKTIV:2	"0" or "1"	F	Automatic failure entry	If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper action limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.OWG:2	N12.4		Upper warning limit of the second control chart	
*.OWGAKTIV:2	"0" or "1"	F	Automatic failure entry	If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper warning limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.MWAVG:2	N12.4		Mean value of the second (Xq) control chart	
*.UWG:2	N12.4		Lower warning limit of the second control chart	The parameter must be empty for PRUEFTYP<>V .
*.UWGAKTIV:2	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V . If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower warning limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.UEG:2	N12.4		Lower action limit of the second control chart	The parameter must be empty for PRUEFTYP<>V .
*.UEGAKTIV:2	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V . If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower action limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.VORG:2	C50	F	Default value for limit value calculation	The following values are valid: "CPK“, "SIGMA“, "QUER“, “ABW_REL“, "ABW_PROZ“

`*.CPK:2`	N12.4		CPK default value for limit value calculation	See CPK:1 parameters
`*.STATVERT:2`	C50	F	Unilateral or bilateral calculation of limit values	See STATVERT:1 parameters
`*.EWEG:2`	C50	F	Non-action probability for the action limits	See EWEG:1 parameters
`*.EWWG:2`	C50	F	Non-action probability for warning limits	See EWEG:1 parameters
`*.RELEG:2`	N12.4		relative deviation/ deviation in percent of the action limit(s) of the second (Xq) control chart from the target value	The parameter must be empty for KARTE:2<>XQ
`*.RELWG:2`	N12.4		relative deviation/ deviation in percent of the warning limit(s) of the second (Xq) control chart from the target value	The parameter must be empty for KARTE:2<>XQ
`*.QUER:2`	N12.4		Default value for sq: or rather for Rq: for computing limit values	
`*.SIGMA:2`	N12.4		Sigma default value for limit value calculation	
`*.XQVART:2`	C50	F	Type of Xq default value (for limit value calculation)	The following values are valid: "RKXQMITTE", "SOLLWERT", "TOLMITTE", "VORGABE"
`*.VORGXQ:2`	N12.4		Default value Xq for limit value calculation	For VORGXQ:2<>VORGABE, the parameter must be empty.
`*.MOD:BER_WG_2`	"0" or "1"	F	Automatically calculate warning limits for the second control chart	If the value "1" is transferred to this parameter, the warning limits for the second control chart are calculated automatically.
`*.MOD:BER_EG_2`	"0" or "1"	F	Automatically calculate action limits for the second control chart	If the value "1" is transferred to this parameter, the action limits for the second control chart are calculated automatically.

.FU:1 to.FU:5	C50		Direct user fields	
.FU:6 to.FU:10	N9		Direct user fields	
*.FU:11; *.FU:12	N12.9		Direct user fields	
*.FU:13; *.FU:14	Date		Direct user fields	

2.2.10 Specification list

The following dialogs are available for updating the specification list:

- CSPEZL.MODIFY to create or change entries in the specification list
- CSPEZL.INSERT to create entries of the specification list
- CSPEZL.UPDATE to change entries of the specification list
- CSPEZL.DELETE to delete entries in the specification list

The unique key of the specification list is made up of the fields CSPEZL.BER, CSPEZL.CMMNR, CSPEZL.ATK, CSPEZL.ATKIDX, CSPEZL.KDNR, CSPEZL.LIEFNR, CSPEZL.AGNR, CSPEZL.AGBEZ, CSPEZL.MNR, CPEZL.SONDERF and CSPEZL.EINTIDX and all user fields (CSPEZL.FU:1 to CSPEZL.FU:14).

Dialog: CSPEZL.*				
Parameter	**Type**	**Manda-tory**	**Contents**	**Description**
*.SONDERF	"0" or "1"	M/I/U/D	Is the entry in the specification list a special case?	This parameter defines whether the entry is a normal or a special case.
*.EINTIDX	N9	M/I/U/D	Version of the specification list entry.	The version number is only kept for normal cases and is issued automatically when a new entry is created.
*.BER	C10	M/I/U/D	Area for which the specification list entry is valid.	An area with the corresponding area ID must exist in HYDRA.
*.CMMNR	C50	M/I/U/D	Characteristic number	The link to the corresponding inspection plan characteristic is implemented via this parameter.

*.ATK	C50	M/I/U/D	Article number	An article with the corresponding combination of article number and drawing issue number must exist in HYDRA.
*.ATKIDX	C50	M/I/U/D	Drawing issue number of the article	An article with the corresponding combination of article number and drawing issue number must exist in HYDRA. If required, the drawing issue number may be left empty.
*.KDNR	C50	M/I/U/D	Customer number	A customer with the corresponding customer number must exist in HYDRA.
*.LIEFNR	C50	M/I/U/D	Supplier number	A supplier with the corresponding supplier number must exist in HYDRA.
*.AGNR	C50	M/I/U/D	Operation number	
*.AGBEZ	C250	M/I/U/D	Operation designation	
*.MNR	C50	M/I/U/D	Machine number	
*.AKTIV	"0" or "1"		Activation of the specification list entry.	
*.IGN	"0" or "1"	F	Option to possibly ignore the characteristic when creating the inspection step.	If the value "1" is transferred to this parameter, the characteristic is omitted at the time the inspection step is created, even if the characteristic was defined in the corresponding inspection plan.
*.CEINH	C50		Unit of measure or size	A corresponding unit of measure with this identifier must already exist in HYDRA.

`*.FMT`	C250		Measured values format of the single values	Missing or completing zeros to be suppressed are marked by the "#" sign, positions that must definitely be displayed are illustrated with a "0". The decimal point is marked by a dot and the thousands separator is marked by a comma. Example: "#,##0.00##" This parameter is only needed for variable characteristics. "0,015" *1 *2 If this parameter is not transferred, then the default format defined in the op "0,025" *1 *2 In the client, the number of decimal places is output
`*.GWNORM`	C50		Standard	This parameter can be used to define which standard the tolerance limits are based on. A relevant standard must be defined in HYDRA. Standards are: • "ISO_PASS“ • "ISO7168“ • "ISO2768“ • "EN12420“ This parameter is only required for variable characteristics.
`*.GWID`	C50		Standard entry identifier	For some standards, an identifier can be specified for the corresponding entry. Example: "H7" for ISO fit standards This parameter is only needed for variable characteristics. This parameter is only required for variable characteristics.

*.STPRPLAN	C50	M/I	Sampling scheme identifier	The sampling scheme must be defined in HYDRA. Possible values: • "NC" • "100PRO" • "SPC" • "LOS" The "LOS" sampling scheme is only valid in the goods receipt or the goods issue areas, while the sampling scheme "SPC" is only valid in the production areas.
*.STPRUMF	N9	M/I	Sample size	This parameter determines the sample size. The value 0 must be used for open samples. The sampling scheme "100PRO" requires an entry (must not be empty)
*.RWMENGE	N9		Acceptance quantity	For STPRPLAN<>NC, the parameter must be empty
*.RUMENGE	N9		Rejection quantity	For STPRPLAN<>NC, the parameter must be empty
*.INTTYP	C50	F	Type of interval	With STPRPLAN<>SPC or STPRPLAN<>NC, the parameter must be empty. Valid values: • "KEINS" for no interval • "ZEIT" for time intervals • "STCK" for piece intervals • "EINMAL" for "once"
*.INTERVAL	N9		Interval	Interval value. For piece intervals, the number of units is set in this field, for time intervals, the interval from this field results from the connection with the field CSPEZL.INTEINH.

*.INTEINH	C50	F	Type of interval	Specifies for time intervals which unit is found in the field CSPEZL.INTERVAL. Only identifiers defined in HYDRA can be used. By default, these are the following: • "SEK" – Seconds • "MIN" – Minutes • "STD" – Hours • "TAG" – Days • "MON" – Months • "JAH" – Years
*.INT:ALW	"0" or "1"	F	Characteristic becomes due for inspection when output batches are changed	Provided that MPL is in use
*.INT:MSW	"0" or "1"	F	Characteristic becomes due for inspection when machine statuses are changed	
*.INT:MSWMST	C250		Source status, target status or a combination of source and target status triggering inspections when changing the machine status	Separate configurations by comma, no blanks
*.INT: SCHICHTW	"0" or "1"	F	Characteristic becomes due for inspection when the shift is changed	

*.MASSANG	C50	F	Type of defined constructional measures	Specifies the manner in which the constructional measures are defined. Only identifiers defined in HYDRA can be used. By default, these are the following: • "ABSOLUT" – the constructional measures are defined absolutely in the parameters OPG, OTG, UTG and UPG. *.OWG:1 • "RELATIV" - the constructional measures are defined as a relative deviation from the target value in the parameters OPGREL, OTGREL, UTGREL and UPGREL. N12.4 • "PROZENTUAL" - the constructional measures are defined as a deviation in percent from the target value in the parameters OPGREL, OTGREL, UTGREL and UPGREL.
*.OPG	N12.4		absolute, upper plausibility limit	The parameter must be empty for attributive characteristics or MASSANG<>ABSOLUT or OPGEXT=1.
*.OPGREL	N12.4		relative/ percentage, upper plausibility limit	The parameter must be empty for attributive characteristics or MASSANG=ABSOLUT
*.OTG	N12.4		absolute, upper tolerance limit	The parameter must be empty for attributive characteristics or MASSANG<>ABSOLUT or OPGEXT=1.
*.OTGREL	N12.4		relative/percentage, upper tolerance limit	The parameter must be empty for attributive characteristics or MASSANG=ABSOLUT
*.OTGAKTIV	"0" or "1"	F	Automatic failure entry	The parameter must be empty for attributive characteristics. If the value "1" is transferred to this parameter, a failure entry is automatically generated if single values violate the tolerance limit.
*.SW	N12.4		Target value	The parameter must be empty for attributive characteristics

*.UTG	N12.4		absolute, lower tolerance limit	The parameter must be empty for attributive characteristics or MASSANG<>ABSOLUT or OPGEXT=1.
*.UTGREL	N12.4		relative/ percentage, lower tolerance limit	The parameter must be empty for attributive characteristics or MASSANG=ABSOLUT
*.UTGAKTIV	"0" or "1"	F	Automatic failure entry	The parameter must be empty for attributive characteristics. If the value "1" is transferred to this parameter, a failure entry is automatically generated if single values violate the tolerance limit.
*.UPG	N12.4		absolute, lower plausibility limit	The parameter must be empty for attributive characteristics or MASSANG<>ABSOLUT or OPGEXT=1.
*.UPGREL	N12.4		relative/ percentage, lower plausibility limit	The parameter must be empty for attributive characteristics or MASSANG=ABSOLUT
*.KARTE:1	C50		Identifier for the first control chart	The following values are valid for variable characteristics: "XQ", "R", "S", "X" or "X_MED" The following values are valid for attributive characteristics: "P" or "U"
*.OEG:1	N12.4		upper action limit of the first control chart	
*.OEGAKTIV:1	"0" or "1"	F	Automatic failure entry	If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper action limit is violated by the statistical value of control chart 1 (e.g. Xq).
*.OWG:1	N12.4		upper warning limit of the first control chart	
*.OWGAKTIV:1	"0" or "1"	F	Automatic failure entry	If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper warning limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).

*.MWAVG:1	N12.4		Mean value for the first control chart	
*.UWG:1	N12.4		lower warning limit of the first control chart	The parameter must be empty for attributive characteristics
*.UWGAKTIV:1	"0" or "1"	F	Automatic failure entry	The parameter must be empty for attributive characteristics. If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower warning limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).
*.UEG:1	N12.4		lower action limit of the first control chart	The parameter must be empty for attributive characteristics
*.UEGAKTIV:1	"0" or "1"	F	Automatic failure entry	The parameter must be empty for attributive characteristics. If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower action limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).
*.VORG:1	C50		Default value for limit value calculation	The following values are valid: "CPK", "SIGMA", "QUER", "ABW_REL", "ABW_PROZ"
*.CPK:1	N12.4		CPK default value for limit value calculation	To calculate Sigma from the defined cpk, the upper and lower tolerance limits must be specified and must be distinguishable.

*.STATVERT:1	C50	F	Unilateral or bilateral calculation of limit values	The following values are valid: "EINS", "ZWEI". If this parameter has the value "EINS", the calculation is only performed for either the upper or the lower action/ warning limits. This setting is mandatory for attributive control charts. For R and s control charts, the upper limit values are calculated in this case. For Xq control charts, the calculation performed depends on which tolerance limit is specified. If the upper and lower tolerance limit is specified, the upper and lower warning/ action limits are calculated, even though "unilateral" was selected. If this parameter has the value "ZWEI", the calculation is performed for the upper and lower action/ warning limits.
*.EWEG:1	C50	F	Non-action probability for the action limits	The following values are valid for unilateral control charts: "1", "1.28", "1.64", "1.96", "2", "2.33", "2.58", "3", "3.09", "3.72", "4" The following values are valid for bilateral control charts: "1", "1.28", "1.64", "1.96", "2", "2.28", "2.33", "2.58", "3", "3.09", "3.45", "3.72", "3.9", "4" The following values are valid for R and s charts: "0.9", "0.95", "0.99"
*.EWWG:1	C50	F	Non-action probability for warning limits	See EWEG:1 parameters
*.RELEG:1	N12.4		relative deviation/ deviation in percent of the action limit(s) of the first (Xq) control chart from the target value	The parameter must be empty for KARTE:1<>XQ

`*.RELWG:1`	N12.4		relative deviation/ deviation in percent of the warning limit(s) of the first (Xq) control chart from the target value	The parameter must be empty for KARTE:1<>XQ
`*.QUER:1`	N12.4		Default value for sq: or rather for Rq: for computing limit values	
`*.SIGMA:1`	N12.4		Sigma default value for limit value calculation	
`*.XQVART:1`	C50	F	Type of Xq default value (for limit value calculation)	The following values are valid: "RKXQMITTE", "SOLLWERT", "TOLMITTE", "VORGABE"
`*.VORGXQ:1`	N12.4		Default value Xq for limit value calculation	For VORGXQ:1<>VORGABE, the parameter must be empty.
`*.MOD:BER_WG_1`	"0" or "1"	F	Automatically calculate warning limits for the first control chart	If the value "1" is transferred to this parameter, the warning limits for the first control chart are calculated automatically. It is not possible to calculate limit values for the Median chart.
`*.MOD:BER_EG_1`	"0" or "1"	F	Automatically calculate action limits for the first control chart	If the value "1" is transferred to this parameter, the action limits for the first control chart are calculated automatically. It is not possible to calculate limit values for the Median chart.
`*.KARTE:2`	C50	F	Identifier of the second control chart	The following values are valid for variable characteristics: "XQ", "R", "S", "X" or "X_MED" The following values are valid for attributive characteristics: "P" or "U"
`*.OEG:2`	N12.4		upper action limit of the second control chart	
`*.OEGAKTIV:2`	"0" or "1"	F	Automatic failure entry	If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper action limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).

*.OWG:2	N12.4		Upper warning limit of the second control chart	
*.OWGAKTIV:2	"0" or "1"	F	Automatic failure entry	If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper warning limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.MWAVG:2	N12.4		Mean value for the second control chart	
*.UWG:2	N12.4		Lower warning limit of the second control chart	The parameter must be empty for attributive characteristics
*.UWGAKTIV:2	"0" or "1"	F	Automatic failure entry	The parameter must be empty for attributive characteristics. If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower warning limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.UEG:2	N12.4		Lower action limit of the second control chart	The parameter must be empty for attributive characteristics
*.UEGAKTIV:2	"0" or "1"	F	Automatic failure entry	The parameter must be empty for attributive characteristics. If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower action limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.VORG:2	C50	F	Default value for limit value calculation	The following values are valid: "CPK", "SIGMA", "QUER", "ABW_REL", "ABW_PROZ"
*.CPK:2	N12.4		CPK default value for limit value calculation	See CPK:1 parameters
*.STATVERT:2	C50	F	Unilateral or bilateral calculation of limit values	See STATVERT:1 parameters
*.EWEG:2	C50	F	Non-action probability for the action limits	See EWEG:1 parameters

*.EWWG:2	C50	F	Non-action probability for warning limits	See EWEG:1 parameters
*.RELEG:2	N12.4		relative deviation/ deviation in percent of the action limit(s) of the second (Xq) control chart from the target value	The parameter must be empty for KARTE:2<>XQ
*.RELWG:2	N12.4		relative deviation/ deviation in percent of the warning limit(s) of the second (Xq) control chart from the target value	The parameter must be empty for KARTE:2<>XQ
*.QUER:2	N12.4		Default value for sq: or rather for Rq: for computing limit values	
*.SIGMA:2	N12.4		Sigma default value for limit value calculation	
*.XQVART:2	C50	F	Type of Xq default value (for limit value calculation)	The following values are valid: "RKXQMITTE", "SOLLWERT", "TOLMITTE", "VORGABE"
*.VORGXQ:2	N12.4		Default value Xq for limit value calculation	For VORGXQ:2<>VORGABE, the parameter must be empty.
*.MOD:BER_WG_2	"0" or "1"	F	Automatically calculate warning limits for the second control chart	If the value "1" is transferred to this parameter, the warning limits for the second control chart are calculated automatically. It is not possible to calculate limit values for the Median chart.
*.MOD:BER_EG_2	"0" or "1"	F	Automatically calculate action limits for the second control chart	If the value "1" is transferred to this parameter, the action limits for the second control chart are calculated automatically. It is not possible to calculate limit values for the Median chart.
.FU:1 to.FU:5	C50		Direct user fields	
.FU:6 to.FU:10	N9		Direct user fields	
*.FU:11; *.FU:12	N12.9		Direct user fields	
*.FU:13; *.FU:14	Date		Direct user fields	

2.2.11 Inspection plans

The following dialogs are available for updating inspection plans:

- `CPPL.IFCMARK` to mark inspection plan data prior to modifying them and to then delete any sub-level data that was not modified (inspection plan criteria)
- `CPPL.MODIFY` to create or modify inspection plan header data
  Only the data in the header of the inspection plan are updated. Inspection plan characteristics are modified using the dialog CPPLMM.
- `CPPL.INSERT` to create inspection plan header data
  Only the data in the header of the inspection plan are updated. Inspection plan characteristics are modified using the dialog CPPLMM.
- `CPPL.UPDATE` to modify inspection plan header data
  Only the data in the header of the inspection plan are updated. Inspection plan characteristics are modified using the dialog CPPLMM.
- `CPPL.COPY` to copy inspection plans
  (including all sub-level data, such as inspection plan characteristics)
- `CPPL.DELETE` to delete inspection plans
  (including all sub-level data, such as inspection plan characteristics)

The inspection plan's unique key is made up of the fields `CPPL.RECTYP`, `CPPL.BER`, `CPPL.PPLID` and `CPPL.PPLIDX`.
To illustrate the user fields completed in the interface in HYDRA, the user fields must have been configured accordingly (object = "CPPL"; user field key = CPPL:RECTYP).

WARNING! HYDRA-CAQ provides the option to copy the user fields of the inspection plan to the inspection requirement when you create an inspection requirement. This functionality is controlled in an option (option 1058). If this functionality is activated, you must guarantee that the numbers of the user fields of inspection plan and inspection requirement are identical.

Example: The interface populates the user fields CPPL.FU:1 to CPPL:FU:14 of the inspection plan header. User fields CPAN.FU:1 to CPAN.FU:14 are passed to the inspection requirements. As a result, user fields 1 to 14 are shown in an inspection requirement based on an inspection plan.
In this case, when configuring the user fields, make sure that the field types for the user fields 1 to 14 in both the inspection plans and in the inspection requirements match.

2.2.11.1 General notes about processing inspection plan data

- If an inspection plan has already been used in HYDRA to create an inspection requirement, it may no longer be edited or deleted. This applies to the same extent to the corresponding inspection plan characteristics and documents.
  This function cannot be bypassed. In a case such as this, an inspection plan must be created (for example with a new inspection plan index). If necessary, this process can be automated using inspection plan version management.

- If an inspection plan was activated (CPPL.AKTIV=1), it may also not be modified. Also, no inspection plan characteristics and documents of this inspection plan may be modified either.
  However, this function can be bypassed with the parameter CPPL.NOACTIVECHK=1.
  If necessary, inspection plan version management can be used to copy the current inspection plan. Changes are then made to the copy.

- An inspection plan can only be activated (CPPL.AKTIV=1) if it has been released (CPPL.FREI=1).
  However, this function can be bypassed with the parameter CPPL.NOACTIVECHK=1.

- An inspection plan can only be released (CPPL.FREI=1) if at least one inspection plan characteristic has been assigned to it.
  However, this function can be bypassed with the parameter CPPL.NOACTIVECHK=1.

- If the system has been appropriately configured, as soon as one inspection plan is activated, all other inspection plans with the same header data (RECTYP, BER, ATK, ATKIDX, KDNR, LIEFNR, HERSTNR, AGNR, AGBEZ, PPLAGCFG) will be deactivated.

2.2.11.2 Inspection plan version management

The inspection plan version management is an option allowing the inspection plan update via an interface to be automated and to automatically manage the versions of an inspection plan. Version management can be activated with the dialog *.MODIFY for inspection plans, inspection plan characteristics and the corresponding documents using the parameter *.MOD:VERSMAN=1.

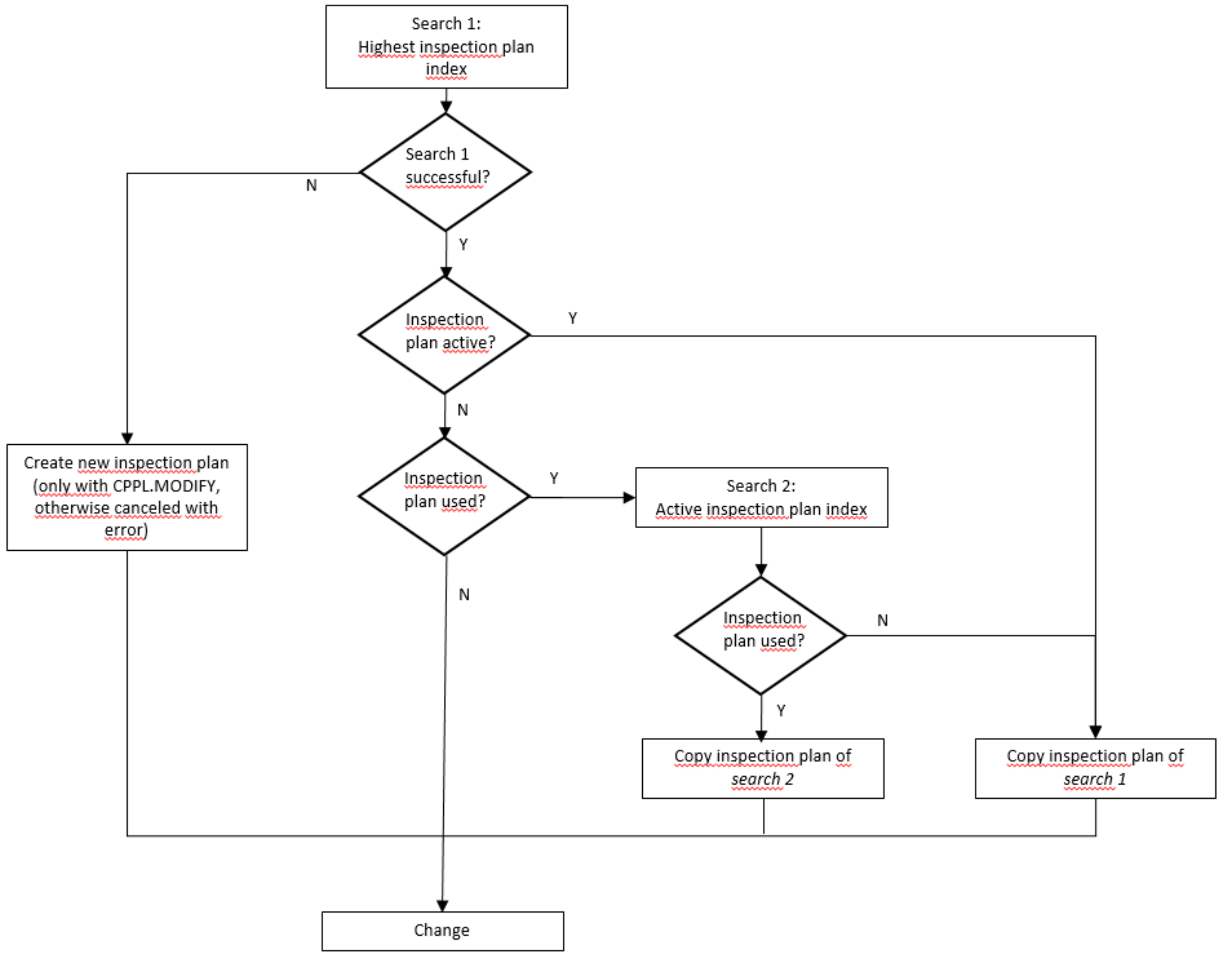
Version management can only be used if the inspection plan index has a standard structure. This structure is described by HYDRA-CAQ option 1053. In the search for the highest inspection plan index, only those inspection plans are considered with an inspection plan index structure matching the sample of option 1053. If this option is not defined or its content is empty, then version management is not used.

In order to identify the inspection plan, only the parameters *.RECTYP, *.BER and *.PPLID must be transferred. The current inspection plan version is calculated automatically, which is why the parameters *.PPLIDX and *.PPLNR are not required.
If, however, they are transferred, version management is not used.

If no inspection plan was found to be copied, then a new inspection plan is created. To create a new inspection plan, you require the CPPL.MODIFY dialog, because the header data of the inspection plan being created is not known for the inspection plan characteristics and the inspection plan documents.
In this case, the dialogs CPPLMM.MODIFY and CPPLDOK.MODIFY are canceled and an error is issued. In the following examples, a sequence is shown that avoids problems at this point with a CPPL.MODIFY that was executed beforehand.

If the current inspection plan is already activated or if it is used in an inspection requirement, it is copied and the changes are made to the copy. This copy is given a new, the next highest, inspection plan index. Release and activation are reset in the new version. The copied inspection plan can then be released/activated manually or, after all changes have been made, this can be performed by the interface.

The functions described above are illustrated in the following flow chart:



2.2.11.3 Example of importing inspection plan data once (safe method)

1. In this example, inspection plans are created by first writing header data. However, these are created without a release and without an activation

```
DLG=CPPL.MODIFY|…|CPPL.FREI=0|CPPL.AKTIV=0|…|CPPL.NOACTIVECHK=0
```

2. Then, all of the characteristics and documents that belong to the inspection plan are created.

```
DLG=CPPLDOK.MODIFY|…|CPPLDOK.NOACTIVECHK=0
DLG=CPPLMM.MODIFY|…|CPPLMM.AFO=10|…|CPPLMM.NOACTIVECHK=0
DLG=CPPLDOK.MODIFY|…|CPPLDOK.AFO=10|…|CPPLDOK.NOACTIVECHK=0
DLG=CPPLMM.MODIFY|…|CPPLMM.AFO=20|…|CPPLMM.NOACTIVECHK=0
DLG=CPPLDOK.MODIFY|…|CPPLDOK.AFO=20|…|CPPLDOK.NOACTIVECHK=0
          :                      :
```

3. Finally, the inspection plan that is now complete can be released and activated. This step, of course, is optional and is only advisable provided that no inspection plans were manually activated.

```
DLG=CPPL.MODIFY|…|CPPL.FREI=1|CPPL.AKTIV=1|…|CPPL.NOACTIVECHK=0
```

2.2.11.4 Example of importing inspection plan data once (quick method)

1. In this example, inspection plans are created by creating header data first. Any release or activation is made in this case immediately. To prevent potential error messages, the parameter `CPPL.NOACTIVECHK=1` is set.

```
DLG=CPPL.MODIFY|…|CPPL.FREI=1|CPPL.AKTIV=1|…|CPPL.NOACTIVECHK=1
```

2. Then, all of the characteristics and documents that belong to the inspection plan are created. In this case as well, any error message is suppressed with the parameter `CPPL.NOACTIVECHK=1`.

```
DLG=CPPLDOK.MODIFY|…|CPPLDOK.NOACTIVECHK=1
DLG=CPPLMM.MODIFY|…|CPPLMM.AFO=10|…|CPPLMM.NOACTIVECHK=1
DLG=CPPLDOK.MODIFY|…|CPPLDOK.AFO=10|…|CPPLDOK.NOACTIVECHK=1
DLG=CPPLMM.MODIFY|…|CPPLMM.AFO=20|…|CPPLMM.NOACTIVECHK=1
DLG=CPPLDOK.MODIFY|…|CPPLDOK.AFO=20|…|CPPLDOK.NOACTIVECHK=1
        :                      :
```

WARNING!! By suppressing error messages, what may happen when importing inspection plan data this way is that released and active inspection plans exist without characteristics.

2.2.11.5 Example of a permanent inspection plan interface with version management

For the example described below, to identify the inspection plan, only the parameters `*.RECTYP`, `*.BER` and `*.PPLID` are required. The parameter `*.PPLIDX` can be omitted by using version management. In addition, the parameter `*.NOACTIVECHK=0` must be transferred in all dialogs.

1. First, inspection plans are created or modified by writing header data. However, they are not released or activated. This step is required in order to configure the necessary header data in the event that a new inspection plan needs to be created (if a search for a current version fails).

```
DLG=CPPL.MODIFY|…|CPPL.FREI=0|CPPL.AKTIV=0|…|CPPL.MOD:VERSMAN=1
```

2. Then, all of the characteristics and documents that belong to the inspection plan are created or modified.

```
DLG=CPPLDOK.MODIFY|…|CPPLDOK.NOACTIVECHK=0
DLG=CPPLMM.MODIFY|…|CPPLMM.AFO=10|…|CPPLMM.MOD:VERSMAN=1
DLG=CPPLDOK.MODIFY|…|CPPLDOK.AFO=10|…|CPPLDOK.MOD:VERSMAN=1
DLG=CPPLMM.MODIFY|…|CPPLMM.AFO=20|…|CPPLMM.MOD:VERSMAN=1
DLG=CPPLDOK.MODIFY|…|CPPLDOK.AFO=20|…|CPPLDOK.MOD:VERSMAN=1
        :                      :
```

3. Finally, the inspection plan that is now complete can be released and activated. This step, of course, is optional and is only advisable provided that no inspection plans were manually activated.

```
DLG=CPPL.MODIFY|…|CPPL.FREI=1|CPPL.AKTIV=1|…|CPPL.MOD:VERSMAN=1
```

2.2.11.6 Example of a permanent inspection plan interface without version management

1. In this example, all of the data assigned to the inspection plan (characteristics and documents) are highlighted first.

```
DLG=CPPL.IFCMARK|CPPL.MOD=MARK|…
```

2. Then, the inspection plan header data is modified. In the process, release and activation can be set at the same time. To prevent any error messages, the parameter `CPPL.NOACTIVECHK=1` is set.

```
DLG=CPPL.MODIFY|…|CPPL.FREI=1|CPPL.AKTIV=1|…|CPPL.NOACTIVECHK=1
```

3. Now, all of the characteristics and documents that belong to the inspection plan are created or modified. What is important for this method is that all characteristics are always written, even those that were not modified. Any error message is again suppressed with the parameter `CPPL.NOACTIVECHK=1`.

```
DLG=CPPLDOK.MODIFY|…|CPPLDOK.NOACTIVECHK=1
DLG=CPPLMM.MODIFY|…|CPPLMM.AFO=10|…|CPPLMM.NOACTIVECHK=1
DLG=CPPLDOK.MODIFY|…|CPPLDOK.AFO=10|…|CPPLDOK.NOACTIVECHK=1
DLG=CPPLMM.MODIFY|…|CPPLMM.AFO=20|…|CPPLMM.NOACTIVECHK=1
DLG=CPPLDOK.MODIFY|…|CPPLDOK.AFO=20|…|CPPLDOK.NOACTIVECHK=1
          :                          :
```

4. In the last step, all of the sub-data that has not been edited (characteristics and documents) are deleted.

```
DLG=CPPL.IFCMARK|CPPL.MOD=DELMARK|…
```

WARNING!! In step 4 of this method, any documents that were created manually are deleted as well.

By suppressing error messages, what may happen when importing inspection plan data this way is that released and active inspection plans exist without characteristics.

2.2.11.7 Field definitions

Dialog: `CPPL.*`				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
`*.RECTYP`	"FEP", "WEP", "WAP", "EMU"	K/M/I/U/C/D	Data type of inspection plan	FEP = In-production inspection plan WEP = Goods receipt inspection WAP = Goods issue inspection plan EMU = Initial sample inspection plan

`*.BER`	C10	K/M/I/ U/C/D	Area for which the inspection plan applies.	An area with the corresponding area ID must exist in HYDRA. By default, these are: E = Goods receipt F = Production A = Goods issue EMU = Initial sample
`*.PPLID`	C50	K/M/I/ U/C/D	Inspection plan number	
`*.PPLIDX`	C50	K/(M)/ (I)/(U) /C/D	Inspection plan index	For CPPL.MODIFY only the mandatory field if the parameter is CPPL.MOD:VERSMAN=0.
`*.RECTYP:Z`	"FEP", "WEP", "WAP", "EMU"	C	Data type of inspection plan (copy destination)	This parameter is only needed for CPPL.COPY. FEP = In-production inspection plan WEP = Goods receipt inspection WAP = Goods issue inspection plan EMU = Initial sample inspection plan Inspection plans may only be copied within one data type in order to avoid problems with changing types of data collection.
`*.BER:Z`	C10	C	Area for which the inspection plan applies (copy destination).	This parameter is only needed for CPPL.COPY. An area with the corresponding area ID must exist in HYDRA.
`*.PPLID:Z`	C50	C	Inspection plan number (copy destination)	This parameter is only needed for CPPL.COPY.
`*.PPLIDX:Z`	C50	C	Inspection plan index (copy destination)	This parameter is only needed for CPPL.COPY.

*.MOD	C50	K	CPPL.IFCMARK method mode	This parameter is only needed for CPPL.IFCMARK. This parameter determines how the dialog functions. The following values are available: "MARK" – all of the data assigned to the inspection plan (characteristics and documents) are highlighted "UNMARK" – highlighting is removed "DELMARK" – all of the data still highlighted (characteristics and documents) are deleted
*.NOACTIVECHK	"0" or "1"	F	Parameter that determines whether a check should run after the inspection plan is released or activated.	If this parameter has the value "1", no check for release or for activation is run. Otherwise, an error message is issued if an active inspection plan should be modified or if an inspection plan that has not been released should be activated.
*.MOD:VERSMAN	"0" or "1"	M/I/U	Parameter specifying whether inspection plan version management should be activated.	This parameter is only needed for CPPL.MODIFY. If this parameter has the value "1", inspection plan version management is activated.
*.ATK	C50		Article number	An article with the corresponding combination of article number and drawing issue number must exist in HYDRA. In the event of an inspection plan for an article group, the number of the article group is entered here (see section "article catalog")

`*.ATKIDX`	C50		Drawing issue number of the article	An article with the corresponding combination of article number and drawing issue number must exist in HYDRA. If required, the drawing issue number may be left empty. In the event of an inspection plan for the article group, the drawing issue number must be left empty.
`*.KDNR`	C50		Customer number	For RECTYP=WEP, the parameter must be empty. A customer with the corresponding customer number must exist in HYDRA.
`*.HERSTNR`	C50		Manufacturer number	For RECTYP<>WEP, the parameter must be empty. A manufacturer with the corresponding manufacturer number must exist in HYDRA.
`*.LIEFNR`	C50		Supplier number	A supplier with the corresponding supplier number must exist in HYDRA.
`*.AGNR`	C50		Operation number	For RECTYP=EMU, the parameter must be empty.
`*.AGBEZ`	C250		Operation designation/name	For RECTYP=EMU, the parameter must be empty.
`*.ZEICHNNR`	C250		Drawing number	

*.EMUFORM	C50	F	Identifier of the corresponding EMU form	For RECTYP<>EMU, the parameter must be empty. An EMU form with the corresponding abbreviation must be defined in HYDRA (status type "EMUFORM"). The default types are listed below: "VDA24"
*.FREI	"0" or "1"	F	Inspection plan release	
*.FREIDAT	Date		Date of inspection plan release	
*.FREIVON	C50		Person who released the inspection plan	
*.AKTIV	"0" or "1"	F	Activate inspection plan	
*.GUELTVON	Date		Date of validity from	
*.GUELTBIS	Date		Date of validity until	
*.PPLAGCFG	C50	F	Parameter specifying at which level operations are defined	For RECTYP=EMU the parameter must always have the value PPL1AG. The parameter used here must be defined in HYDRA. Possible values: • "PPL1AG" – for each operation there is exactly one inspection plan • "PPLxAG" – there is an inspection plan for all operations For the default configuration, the setting "PPLxAG" is to be used as the default configuration. Parameter "PPL1AG" must not be used without prior consultation.

*.QMAGCR	C50		Generate operations, if required	• "NONE" - When generating new inspection requirements the structure of working plans is not changed and no new "true" QM OPs are created. • "PAN_CREATE" – When generating inspection requirements, new true QM OPs are created, if needed.
*.MERKAKT	C50		Generation of inspection steps and characteristics	• "PAN_CREATE" – Corresponding inspection steps and characteristics are assigned to all operations upon generation of an inspection requirement. • "AG_AN" – Inspection steps and corresponding characteristics are generated when logging on an operation. This configuration is only important for cases when inspection requirements and corresponding inspection steps are created because operations are logged on or order statuses are changed. This parameter is irrelevant for the direct generation of inspection requirements (manually or via the interface). As in this case, characteristics are always created along with the generation of inspection requirements.
*.PAUPPLATZ	C50	F	Parameter used for splitting inspection steps by inspection stations	The parameter used here must be defined in HYDRA. Possible values: • "PAU1PP" – one inspection step is generated for all inspection stations • "PAUxPP" – one inspection step is generated for each inspection station

*.NESTZUORD	C50	F	Parameter used to assign cavity information	For RECTYP=EMU the parameter must always have the value KEINE. The parameter used here must be defined in HYDRA. Possible values: • "KEINE" – no cavities are assigned • "STICHPROBE" – the cavities are assigned to the sample This value must only be used if inspection points are in use (by default with CPPL.RECTP=FEP)
*.PAUAKTION	C50	F	Parameter for releasing inspection steps	The parameter used here must be defined in HYDRA. Possible values: • "PAUERST" – the inspection steps are created, but not released • "PAUFREI" – the inspection steps are created and immediately released
*.PRUEFART	C50	F	Type of characteristic change during the inspection	The parameter used here must be defined in HYDRA. Possible values: • "MERKMAL" – the inspection is performed by characteristic • "STUECK" – piece-related inspection is done This value must not be used together with CPPL.NESTZUORD=STICHPROBE
*.DYNART	C50	F	Type of dynamic modification	This parameter must be empty for data types that are not dynamic (by default: FEP, EMU). The parameter entered here must be defined in HYDRA. Possible values:

				• "KEINE" – nothing is dynamically modified • "LOS" – dynamic modification is performed at batch level (inspection requirements) • "MERKMAL" – dynamic modification is performed at characteristics level
`*.DYUEBERG`	C50	F	Transitional definition for dynamic modification	For DYNART<>LOS, the parameter must be empty. The transitional definition entered here must be defined in HYDRA. Default value is "DIN_ISO".
`*.DYPSCHARF:STA`	C50	F	Initial inspection severity for dynamic modification	For DYNART<>LOS, the parameter must be empty. The inspection severity entered here must be known in the inspection severity definition assigned to the transitional definition.
`*.FU:1` to `*.FU:5`	C50		Direct user fields	
`*.FU:6` to `*.FU:10`	N9		Direct user fields	
`*.FU:11`; `*.FU:12`	N12.9		Direct user fields	
`*.FU:13`; `*.FU:14`	Date		Direct user fields	

2.2.12 Inspection plan characteristics

The following dialogs are available for updating inspection plans:

- `CPPLMM.MODIFY` to create or change inspection plan characteristics
- `CPPLMM.INSERT` to create inspection plan characteristics
- `CPPLMM.UPDATE` to change inspection plan characteristics
- `CPPLMM.DELETE` to delete inspection plan characteristics

The inspection plan's unique key is made up of the fields `CPPLMM.RECTYP`, `CPPLMM.BER`, `CPPLMM.PPLID`, `CPPLMM.PPLIDX` and `CPPLMM.AFO`.
Before creating or changing inspection plan characteristics, you must make sure that the corresponding inspection plan already exists.

Dialog: CPPLMM.*				
Parameter	Type	Mandatory	Contents	Description
`*.RECTYP`	"FEP", "WEP", "WAP", "EMU"	M/I/U/D	Data type of inspection plan	FEP = In-production inspection plan WEP = Goods receipt inspection WAP = Goods issue inspection plan EMU = Initial sample inspection plan
`*.BER`	C10	M/I/U/D	Area for which the inspection plan applies.	An area with the corresponding area ID must exist in HYDRA.
`*.PPLID`	C50	M/I/U/D	Inspection plan number	
`*.PPLIDX`	C50	(M)/ (I)/ (U)/D	Inspection plan index	For CPPLMM.MODIFY only the mandatory field if the parameter is CPPLMM.MOD:VERSMAN=0.
`*.AFO`	N9	M/I/U/D	Work sequence (operation sequence) of the characteristic	The work sequence number determines the sequence of the characteristics. It must be unique in the inspection plan.
`*.NOACTIVECHK`	"0" or "1"	M/I	Parameter specifying whether a check should run after the inspection plan is activated.	If this parameter has the value "1", no check for activation is run. Otherwise, an error message is issued if an active inspection plan should be modified.
`*.MOD:VERSMAN`	"0" or "1"	M/I/U	Parameter specifying whether inspection plan version management should be activated.	This parameter is only needed for CPPLMM.MODIFY. If this parameter has the value "1", inspection plan version management is activated.
`*.AGNR`	C50		Operation number	For RECTYP=EMU, the parameter must always be empty.

`*.AGBEZ`	C250		Operation designation/name	For RECTYP=EMU, the parameter must always be empty.
`*.CMMNR`	C50		Characteristic number	If this parameter is specified, a characteristic with the same number must exist in the characteristics catalog.
`*.MMBEZ`	C250		Characteristic designation/name	The characteristic designation/name may be different from the designation/name in the characteristics catalog.
`*.PPLATZ`	C50		Inspection station	If this parameter is specified, a respective inspection station must be defined in HYDRA.
`*.MMTYP`	C50	F	Characteristic type	For RECTYP=EMU the parameter must always have the value PRODUKT. The parameter used here must be defined in HYDRA. Possible values: • "PRODUKT" – Product characteristic • "PROZESS" – Process characteristic
`*.ERFART`	"MANUELL"	F	Fixed identifier for CAQ characteristics	
`*.MUSSPRF`	"0" or "1"	F	Mandatory inspection	For RECTYP=EMU the parameter must always have the value 0. If this parameter has the value 1, an inspection step cannot be completed until at least one measured value was recorded for this characteristic.

*.BFORMEL	C250		Formula	If this parameter is completed with a value, then the characteristic's values will be calculated based on the formula defined here. In this case, it might not be possible to enter measured values for this characteristic manually (depends on the type of formula). Refer to the corresponding documentation to see the structure of a formula of this kind.
*.QDETAIL	C50	F	Source of the characteristic detail	For RECTYP=EMU the parameter must always have the value PPLMER. The parameter used here must be defined in HYDRA. Possible values: • "PPLMER" – Details are defined here in the inspection plan characteristic • "MERKAT" – Details are taken from the characteristics catalog (to do this, the parameter CMMNR must be completed)
*.QSPEZ	C50	F	Source of the specifications	For RECTYP=EMU the parameter must always have the value PPLMER. The parameter used here must be defined in HYDRA. Possible values: • "PPLMER" – Specifications are defined here in the inspection plan characteristic • "MERKAT" – Specifications are pulled from the characteristics catalog (to do this, the parameter CMMNR must be filled) • "LISTE" – Specifications are taken from the specification list (to do this, the parameter CMMNR must be filled)

*.PRUEFTYP	C50	F	Type of data collection	For QDETAIL<>PPLMER, the parameter must be empty The parameter used here must be defined in HYDRA. Possible values: • "A" - attributive • "V" - variable • "F" – inspection chart (chart of recorded defects)
*.ERFASSDET	C50		Kind of input (inspection) type	This parameter is only available as of CAQ 8.2. The parameter used here must be defined in HYDRA. Possible values: • "“ – Standard • "RASTER“ – Visual defects recording → only supported with CPPLMM.PRUEFTYP=F • "CODE“ – based on catalogs → only supported with CPPLMM.PRUEFTYP=A • "CODE_ZUFALL“ – based on catalogs (random) → only supported with CPPLMM.PRUEFTYP=A
*.BEWKAUSWMEN:1	C10		Selected set for assessment catalog	This parameter is only available as of CAQ 8.2. This parameter is only supported with CPPLMM.ERFASSDET=CODE or CPPLMM.ERFASSDET=CODE_ZUFALL.
*.PRFRASTER:X	C250		Grid of x-axis	This parameter is only available as of CAQ 8.2. This parameter is only supported with CPPLMM.ERFASSDET=RASTER The identifiers of the grid must be separated by comma.

*.PRFRASTER:Y	C250		Grid of y-axis	This parameter is only available as of CAQ 8.2. This parameter is only supported with CPPLMM.ERFASSDET=RASTER The identifiers of the grid must be separated by comma.
*.PMID	C50		Type PRM (test equipment/gage) resource to be used	For QDETAIL<>PPLMER, the parameter must be empty If this parameter is specified, a respective PRM (test equipment/gage) type resource must be defined in HYDRA. If this entry is used, the parameter CPPLMM.RESFAM must be empty.
*.RESFAM			Resource family to be used	If this parameter is specified, a respective resource family must be defined in HYDRA. If this entry is used, the parameter CPPLMM.PMID must be empty.
*.OPT:PLAN	M or G	F	M = machine, G = machine group	Defined here is whether the scheduling is made on a machine/ workplace or on a machine/ workplace group. For RECTYP<>EMU, the parameter must always have the value 0. M is the initial assignment If this parameter has the value 1, this characteristic will be transferred into non-EMU inspection plans.
*.MNR	C50		Planning on a machine/ a workplace	If this parameter is specified, a respective workplace must be defined in HYDRA. The QM operation generated for this characteristic can only be logged on to the specified machine and inspected on it.

*.MGRP	C 20		Planning on a machine group/ a workplace group	If this parameter is specified, a respective workplace group must be defined in HYDRA. The QM operation generated for this characteristic can be found in the sequencing list for all machines/ workplaces belonging to this group. As such, it can be inspected at all machines with this group reference.
*.ZERTPRN	C50	F	Certificate printing	For QDETAIL<>PPLMER, the parameter must be empty The ID entered here must be defined in HYDRA. Default values are: • "NIE" – never print • "AUSW" – selectable • "IMMER" – always print
*.ERRGEW	C50	F	Failure weighting	For QDETAIL<>PPLMER, the parameter must be empty The error weighting entered here must be defined in HYDRA. Default values are: • "NEBEN" – Minor defect • "HAUPT" – Major defect • "KRIT" – Critical defect
*.BPRUEFERG	C50		Inspection result base	Specifies the basis for identifying the inspection result of the characteristic. • NCD_ALL = inspection result is calculated over all samples • NCD_LAST = inspection result is calculated from the last sample
*.ANAUSNR	C50		Analysis selection catalog	If this parameter is specified, a respective analysis selection catalog must be defined in HYDRA.

*.KEINNEST	"0" or "1"	F	No cavity-related inspection	For QDETAIL<>PPLMER or RECTYP=EMU, the parameter must have the value 0. If this parameter has the value 1, then this characteristic, as opposed to the settings in the inspection plan header, will not be inspected with regard to cavities.
*.EMUMMTYP	C50	F	Characteristics category for initial sample characteristics.	For RECTYP<>EMU, the parameter must be empty. A characteristics category with the corresponding abbreviation must be defined for the initial sample form specified in the inspection plan header. The standard types for them as per VDA volume 2, 4th edition, are listed below: • MASS for measurement test • FUNKTION for function test • WERKSTOFF for material test • HAPTIK for haptic test • AKUSTIK for acoustic test • GERUCH for odor test • AUSSEHEN for exterior appearance test / visual test • OBERFLAECHE for surface test • EMV for EMC test • ZUVERLAESSIG for reliability test The categories are EMUTYP_VDA24 type status entries
*.DYNAM	"0" or "1"	F	Will the characteristic be dynamically modified?	For DYNART<>MERKMAL (in the corresponding inspection plan header data), the parameter must have the value 0. If this parameter has the value 1, this characteristic will be dynamically modified.

*.DYUEBERG	C50	F	Transitional definition for dynamic modification	For DYNART<>MERKMAL (in the corresponding inspection plan header data), the parameter must be empty. The transitional definition entered here must be defined in HYDRA. Default value is "DIN_ISO".
*.DYNORM	C50	F	Dynamic modification norm	For DYNART<>KEINE (in the corresponding inspection plan header data), the parameter must be empty. The transitional definition entered here must be defined in HYDRA. Default values are: "ISO_3951" –variable "ISO_2859" - attributive
*.PNIVEAU	C50	F	Inspection level of the dynamic modification norm	For DYNART<>KEINE (in the corresponding inspection plan header data), the parameter must be empty. The inspection level entered here must be defined for the corresponding dynamic modification norm. Default values are: "I" *1 *2 "II" *1 *2 "III" *1 *2 "S-1" *1 "S-2" *1 "S-3" *1 *2 "S-4" *1 *2 *1 – for DYNORM=ISO_2859 *2 – for DYNORM=ISO_3951
*.AQL	C50	F	AQL value of the dynamic modification norm *.ZUSCHR:2	For DYNART<>KEINE (in the corresponding inspection plan header data), the parameter must be empty. The AQL value entered here must be defined for the corresponding dynamic modification norm. Default values are: „0.010“ *1 *2 „0.015“ *1 *2 „0.025“ *1 *2 „0.040“ *1 *2

				„0.065“ *1 *2 „0.10“ *1 „0.15“ *1 „0.25“ *1 „0.40“ *1 „0.65“ *1 „1.0“ *1 *2 „1.5“ *1 *2 „2.5“ *1 *2 „4.0“ *1 *2 „6.5“ *1 *2 „10“ *1 *2 „15“ *1 „25“ *1 „40“ *1 „65“ *1 „100“ *1 „150“ *1 „250“ *1 „400“ *1 „650“ *1 „1000“ *1 *1 – for DYNORM=ISO_2859 *2 – for DYNORM=ISO_3951
*.DYNMETH	C50	F	Dynamic modification norm method	For DYNART<>KEINE (in the corresponding inspection plan header data), the parameter must be empty. The method entered here must be defined for the corresponding dynamic modification norm. Default values are: "s“ *1 "Sigma" *1 *1 – for DYNORM=ISO_3951
*.DYPSCHARF:STA	C50	F	Initial inspection severity for dynamic modification	For DYNART<>KEINE (in the corresponding inspection plan header data), the parameter must be empty. The inspection severity entered here must be known in the inspection severity definition assigned to the transitional definition.

				Default values for the ISO dynamic modification are: "v" – intensified (tightened) inspection "n"– normal inspection "r" – reduced inspection "a" – suspended inspection
`*.CEINH`	C50		Unit of measure or size	For QSPEZ<>PPLMER, the parameter must be empty A corresponding unit of measure with this identifier must already exist in HYDRA.
`*.FMT`	C250		Measured values format of the single values	For QSPEZ<>PPLMER, the parameter must be empty Missing or completing zeros to be suppressed are marked by the "#" sign, positions that must definitely be displayed are illustrated with a "0". The decimal point is marked by a dot and the thousands separator is marked by a comma. Example: "#,##0.00##" This parameter is only needed for variable characteristics. "0,015" *1 *2 If this parameter is not transferred, then the default format defined in the op "0,025" *1 *2 In the client, the number of decimal places is output

*.GWNORM	C50	F	Standard	For QSPEZ<>PPLMER, the parameter must be empty This parameter can be used to define which standard the tolerance limits are based on. A relevant standard must be defined in HYDRA. Standards are: Standards are: • "ISO_PASS" • "ISO7168" • "ISO2768" • "EN12420" This parameter is only required for variable characteristics.
*.GWID	C50	F	Standard entry identifier	For QSPEZ<>PPLMER, the parameter must be empty For some standards, an identifier can be specified for the corresponding entry. Example: "H7" for ISO fit standards This parameter is only required for variable characteristics.
*.STPRPLAN	C50	M/I (see comments)	Sampling scheme identifier	The parameter must be empty for QSPEZ<>PPLMER. The sampling scheme must be defined in HYDRA. Possible values: • "NC" • "100PRO" • "SPC" • "LOS" The "LOS" sampling scheme is only valid in the goods receipt or the goods issue areas, while the sampling scheme "SPC" is only valid in the production areas.

`*.STPRUMF`	N9	M/I (see comments)	Sample size	The parameter must be empty for QSPEZ<>PPLMER. This parameter determines the sample size. The value 0 must be used for open samples. The sampling scheme "100PRO" requires an entry (must not be empty)
`*.RWMENGE`	N9		Acceptance quantity	For QSPEZ<>PPLMER or STPRPLAN<>NC, the parameter must be empty
`*.RUMENGE`	N9		Rejection quantity	For QSPEZ<>PPLMER or STPRPLAN<>NC, the parameter must be empty
`*.INTTYP`	C50	F	Type of interval	With STPRPLAN<>SPC or STPRPLAN<>NC or QSPEZ<>PPLMER, the parameter must be empty. Valid values: • "KEINS" for no interval • "ZEIT" for time intervals • "STCK" for piece intervals • "EINMAL" for "once"
`*.INTERVAL`	N9		Interval	With STPRPLAN<>SPC or STPRPLAN<>NC or QSPEZ<>PPLMER, the parameter must be empty. Interval value. For piece intervals, the number of units is set in this field, for time intervals, the interval from this field results from the connection with the field CSPEZL.INTEINH.

`*.INTEINH`	C50	F	Type of interval	With STPRPLAN<>SPC or STPRPLAN<>NC or QSPEZ<>PPLMER, the parameter must be empty. Specifies for time intervals which unit is found in the field CSPEZL.INTERVAL. Only identifiers defined in HYDRA can be used. By default, these are the following: • "SEK" – Seconds • "MIN" – Minutes • "STD" – Hours • "TAG" – Days • "MON" – Months • "JAH" – Years
`*.INT:ALW`	"0" or "1"	F	Characteristic becomes due for inspection when output batches are changed	Provided that MPL is in use
`*.INT:MSW`	"0" or "1"	F	Characteristic becomes due for inspection when machine statuses are changed	
`*.INT:MSWMST`	C250		Source status, target status or a combination of source and target status triggering inspections when changing the machine status	Separate configurations by comma, no blanks
`*.INT: SCHICHTW`	"0" or "1"	F	Characteristic becomes due for inspection when the shift is changed	
`*.PRBZUG`	"0" or "1"	F	Sampling	This parameter can be used from SP4 onwards.

*.PRBGRP	C50		Sample group	This parameter can be used from SP4 onwards.
*.MASSANG	C50	F	Type of defined constructional measures	For QSPEZ<>PPLMER, the parameter must be empty Specifies the manner in which the constructional measures are defined. Only identifiers defined in HYDRA can be used. By default, these are the following: • "ABSOLUT" – the constructional measures are defined absolutely in the parameters OPG, OTG, UTG and UPG. *.OWG:1 • "RELATIV" - the constructional measures are defined as a relative deviation from the target value in the parameters OPGREL, OTGREL, UTGREL and UPGREL. N12.4 • "PROZENTUAL" - the constructional measures are defined as a deviation in percent from the target value in the parameters OPGREL, OTGREL, UTGREL and UPGREL.
*.OPG	N12.4		absolute, upper plausibility limit	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER or MASSANG<>ABSOLUT .
*.OPGREL	N12.4		relative/ percentage, upper plausibility limit	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER or MASSANG=ABSOLUT .
*.OTG	N12.4		absolute, upper tolerance limit	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER or MASSANG<>ABSOLUT .
*.OTGREL	N12.4		relative/percentage, upper tolerance limit	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER or MASSANG=ABSOLUT .

*.OTGAKTIV	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER. If the value "1" is transferred to this parameter, a failure entry is automatically generated if single values violate the tolerance limit.
*.SW	N12.4		Target value	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER.
*.UTG	N12.4		absolute, lower tolerance limit	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER or MASSANG<>ABSOLUT .
*.UTGREL	N12.4		relative/ percentage, lower tolerance limit	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER or MASSANG=ABSOLUT .
*.UTGAKTIV	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER. If the value "1" is transferred to this parameter, a failure entry is automatically generated if single values violate the tolerance limit.
*.UPG	N12.4		absolute, lower plausibility limit	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER or MASSANG<>ABSOLUT .
*.UPGREL	N12.4		relative/ percentage, lower plausibility limit	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER or MASSANG=ABSOLUT .

*.KARTE:1	C50	F	Identifier for the first control chart	For QSPEZ<>PPLMER, the parameter must be empty The following values are valid for variable characteristics: "XQ", "R", "S", "X" or "X_MED" The following values are valid for attributive characteristics: "P" or "U"
*.OEG:1	N12.4		upper action limit of the first control chart	The parameter must be empty for QSPEZ<>PPLMER .
*.OEGAKTIV:1	"0" or "1"	F	Automatic failure entry	For QSPEZ<>PPLMER, the parameter must be empty If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper action limit is violated by the statistical value of control chart 1 (e.g. Xq).
*.OWG:1	N12.4		upper warning limit of the first control chart	The parameter must be empty for QSPEZ<>PPLMER .
*.OWGAKTIV:1	"0" or "1"	F	Automatic failure entry	For QSPEZ<>PPLMER, the parameter must be empty If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper warning limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).
*.MWAVG:1	N12.4		Mean value for the first control chart	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER .
*.UWG:1	N12.4		lower warning limit of the first control chart	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER .

`*.UWGAKTIV:1`	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER. If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower warning limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).
`*.UEG:1`	N12.4		lower action limit of the first control chart	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER .
`*.UEGAKTIV:1`	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER. If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower action limit is violated by the statistically calculated value of control chart 1 (e.g. Xq).
`*.VORG:1`	C50	F	Default value for limit value calculation	For QSPEZ<>PPLMER, the parameter must be empty The following values are valid: "CPK“, "SIGMA“, "QUER“, “ABW_REL“, "ABW_PROZ“
`*.CPK:1`	N12.4		CPK default value for limit value calculation	For QSPEZ<>PPLMER, the parameter must be empty To calculate Sigma from the defined cpk, the upper and lower tolerance limits must be specified and must be distinguishable.

*.STATVERT:1	C50	F	Unilateral or bilateral calculation of limit values	For QSPEZ<>PPLMER, the parameter must be empty The following values are valid: "EINS", "ZWEI". If this parameter has the value "EINS", the calculation is only performed for either the upper or the lower action/ warning limits. This setting is mandatory for attributive control charts. For R and s control charts, the upper limit values are calculated in this case. For Xq control charts, the calculation performed depends on which tolerance limit is specified. If the upper and lower tolerance limit is specified, the upper and lower warning/ action limits are calculated, even though "unilateral" was selected. If this parameter has the value "ZWEI", the calculation is performed for the upper and lower action/ warning limits.
*.EWEG:1	C50	F	Non-action probability for the action limits	For QSPEZ<>PPLMER, the parameter must be empty The following values are valid for unilateral control charts: "1", "1.28", "1.64", "1.96", "2", "2.33", "2.58", "3", "3.09", "3.72", "4" The following values are valid for bilateral control charts: "1", "1.28", "1.64", "1.96", "2", "2.28", "2.33", "2.58", "3", "3.09", "3.45", "3.72", "3.9", "4" The following values are valid for R and s charts: "0.9", "0.95", "0.99"
*.EWWG:1	C50	F	Non-action probability for warning limits	For QSPEZ<>PPLMER, the parameter must be empty See EWEG:1 parameters

*.RELEG:1	N12.4		relative deviation/ deviation in percent of the action limit(s) of the first (Xq) control chart from the target value	Parameter must be empty for QSPEZ<>PPLMER or KARTE:1<>XQ.
*.RELWG:1	N12.4		relative deviation/ deviation in percent of the warning limit(s) of the first (Xq) control chart from the target value	Parameter must be empty for QSPEZ<>PPLMER or KARTE:1<>XQ.
*.QUER:1	N12.4		Default value for sq: or rather for Rq: for computing limit values	For QSPEZ<>PPLMER, the parameter must be empty
*.SIGMA:1	N12.4		Sigma default value for limit value calculation	For QSPEZ<>PPLMER, the parameter must be empty
*.XQVART:1	C50	F	Type of Xq default value (for limit value calculation)	For QSPEZ<>PPLMER, the parameter must be empty The following values are valid: "RKXQMITTE", "SOLLWERT", "TOLMITTE", "VORGABE"
*.VORGXQ:1	N12.4		Default value Xq for limit value calculation	Parameter must be empty for QSPEZ<>PPLMER or VORGXQ:1<>VORGABE.
*.MOD:BER_WG_1	"0" or "1"	F	Automatically calculate warning limits for the first control chart	If the value "1" is transferred to this parameter, the warning limits for the first control chart are calculated automatically. It is not possible to calculate limit values for the Median chart.

*.MOD:BER_EG_1	"0" or "1"	F	Automatically calculate action limits for the first control chart	If the value "1" is transferred to this parameter, the action limits for the first control chart are calculated automatically. It is not possible to calculate limit values for the Median chart.
*.KARTE:2	C50	F	Identifier of the second control chart	For QSPEZ<>PPLMER, the parameter must be empty The following values are valid for variable characteristics: "XQ", "R", "S", "X" or "X_MED" The following values are valid for attributive characteristics: "P" or "U"
*.OEG:2	N12.4		upper action limit of the second control chart	The parameter must be empty for QSPEZ<>PPLMER .
*.OEGAKTIV:2	"0" or "1"	F	Automatic failure entry	For QSPEZ<>PPLMER, the parameter must be empty If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper action limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.OWG:2	N12.4		Upper warning limit of the second control chart	The parameter must be empty for QSPEZ<>PPLMER .
*.OWGAKTIV:2	"0" or "1"	F	Automatic failure entry	For QSPEZ<>PPLMER, the parameter must be empty If the value "1" is transferred to this parameter, a failure entry is automatically generated if the upper warning limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.MWAVG:2	N12.4		Mean value of the second (Xq) control chart	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER .

*.UWG:2	N12.4		Lower warning limit of the second control chart	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER .
*.UWGAKTIV:2	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER. If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower warning limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.UEG:2	N12.4		Lower action limit of the second control chart	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER .
*.UEGAKTIV:2	"0" or "1"	F	Automatic failure entry	The parameter must be empty for PRUEFTYP<>V or QSPEZ<>PPLMER. If the value "1" is transferred to this parameter, a failure entry is automatically generated if the lower action limit is violated by the statistically calculated value of control chart 2 (e.g. standard deviation).
*.VORG:2	C50	F	Default value for limit value calculation	For QSPEZ<>PPLMER, the parameter must be empty The following values are valid: "CPK“, "SIGMA“, "QUER“, “ABW_REL“, "ABW_PROZ“
*.CPK:2	N12.4		CPK default value for limit value calculation	For QSPEZ<>PPLMER, the parameter must be empty See CPK:1 parameters
*.STATVERT:2	C50	F	Unilateral or bilateral calculation of limit values	For QSPEZ<>PPLMER, the parameter must be empty See STATVERT:1 parameters
*.EWEG:2	C50	F	Non-action probability for the action limits	For QSPEZ<>PPLMER, the parameter must be empty See EWEG:1 parameters

*.EWWG:2	C50	F	Non-action probability for warning limits	For QSPEZ<>PPLMER, the parameter must be empty See EWEG:1 parameters
*.RELEG:2	N12.4		relative deviation/ deviation in percent of the action limit(s) of the second (Xq) control chart from the target value	Parameter must be empty for QSPEZ<>PPLMER or KARTE:2<>XQ.
*.RELWG:2	N12.4		relative deviation/ deviation in percent of the warning limit(s) of the second (Xq) control chart from the target value	Parameter must be empty for QSPEZ<>PPLMER or KARTE:2<>XQ.
*.QUER:2	N12.4		Default value for sq: or rather for Rq: for computing limit values	For QSPEZ<>PPLMER, the parameter must be empty
*.SIGMA:2	N12.4		Sigma default value for limit value calculation	For QSPEZ<>PPLMER, the parameter must be empty
*.XQVART:2	C50	F	Type of Xq default value (for limit value calculation)	For QSPEZ<>PPLMER, the parameter must be empty The following values are valid: "RKXQMITTE", "SOLLWERT", "TOLMITTE", "VORGABE"
*.VORGXQ:2	N12.4		Default value Xq for limit value calculation	Parameter must be empty for QSPEZ<>PPLMER or VORGXQ:2<>VORGABE.
*.MOD:BER_WG_2	"0" or "1"	F	Automatically calculate warning limits for the second control chart	If the value "1" is transferred to this parameter, the warning limits for the second control chart are calculated automatically.

*.MOD:BER_EG_2	"0" or "1"	F	Automatically calculate action limits for the second control chart	If the value "1" is transferred to this parameter, the action limits for the second control chart are calculated automatically.
.FU:1 to.FU:5	C50		Direct user fields	
.FU:6 to.FU:10	N9		Direct user fields	
*.FU:11; *.FU:12	N12.9		Direct user fields	
*.FU:13; *.FU:14	Date		Direct user fields	

2.2.13 Documents of inspection plans/inspection plan characteristics

The following dialogs are available for updating inspection plan documents and/or inspection plan characteristics documents:

- CPPLDOK.MODIFY to create or change document entries
- CPPLDOK.INSERT to create document entries
- CPPLDOK.UPDATE to change document entries
- CPPLDOK.DELETE to delete document entries

If the parameter CPPLDOK.AFO is specified for the dialog, the document entry relates to an inspection plan characteristic. If the parameter is not specified, the document entry relates to the inspection plan header.

The unique key for inspection plan documents is made up of the fields CPPLDOK.RECTYP, CPPLDOK.BER and CPPLDOK.PPLID and CPPLDOK.PPLIDX.

The unique key for inspection plan characteristics documents is made up of the fields CPPLDOK.RECTYP, CPPLDOK.BER, CPPLDOK.PPLID, CPPLDOK.PPLIDX and CPPLDOK.AFO.

Before creating or changing documents, you must make sure that the corresponding inspection plan or inspection plan characteristic already exists.

If the document entry is a "Text" type (CPPLDOK.DOKTYP=TEXT), then the parameters CPPLDOK.TEXT:1 – CPPLDOK.TEXT:10 are used to transfer document texts. The line break (ASCII code 10) is masked with the character string ~"~ (ASCII code: 126, 34, 126).

If the document entry is a "File" type (`CPPLDOK.DOKTYP=DATEI`), then the link to the corresponding file is included in the field `CPPLDOK.DOKURL`. When specifying the file name, this can be done both in UNC (*SERVER\VOLUME\PATH\FILE NAME.EXT*) as well as with the drive reference (*DRIVE:\PATH\FILE NAME.EXT*). If a file should be referenced in a subdirectory of the HYDRA server, this is done by using a forward slash (instead of a backslash and the relative path specification (*./PATH/FILE NAME.EXT*) If the document entry is a "URL" type (`CPPLDOK.DOKTYP=URL`), then the corresponding HTML address is entered in the field `CPPLDOK.DOKURL`.

Dialog: CPPLDOK.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
*.RECTYP	"FEP", "WEP", "WAP", "EMU"	M/I/U/D	Data type of inspection plan	FEP = In-production inspection plan WEP = Goods receipt inspection WAP = Goods issue inspection plan EMU = Initial sample inspection plan
*.BER	C10	M/I/U/D	Area for which the inspection plan applies.	An area with the corresponding area ID must exist in HYDRA.
*.PPLID	C50	M/I/U/D	Inspection plan number	
*.PPLIDX	C50	(M)/(I)/(U)/D	Inspection plan index	For CPPLDOK.MODIFY it is only a mandatory field if the parameter is CPPLDOK.MOD:VERSMAN=0.
*.AFO	N9	(M)/(I)/(U)/(D)	Work sequence (operation sequence) of the characteristic	
*.NOACTIVECHK	"0" or "1"	F	Parameter specifying whether a check should run after the inspection plan is activated.	If this parameter has the value "1", no check for activation is run. Otherwise, an error message is issued if an active inspection plan should be modified.

*.MOD:VERSMAN	"0" or "1"	M/I/U	Parameter specifying whether inspection plan version management should be activated.	This parameter is only needed for CPPLDOK.MODIFY. If this parameter has the value "1", inspection plan version management is activated.
*.DOKNR	N9	M/I/U/D	Document number	This document number must be unique in the inspection plan or in the inspection plan characteristic.
*.DOKBEZ	C250		Document name	
*.DOKTYP	C50	F	Type of document	The document type must be defined in HYDRA. Possible values: • "TEXT" • "DATEI“ • "URL"
*.DOKURL	C250		File name or URL of the document	For DOKTYP=TEXT, the parameter must be empty.
*.TEXT:1	C250		Characters 1-250 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
*.TEXT:2	C250		Characters 251-500 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
*.TEXT:3	C250		Characters 501-750 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
*.TEXT:4	C250		Characters 751-1000 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
*.TEXT:5	C250		Characters 1001-1250 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
*.TEXT:6	C250		Characters 1251-1500 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
*.TEXT:7	C250		Characters 1501-1750 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
*.TEXT:8	C250		Characters 1751-2000 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
*.TEXT:9	C250		Characters 2001-2250 of the document text	For DOKTYP<>TEXT, the parameter must be empty.

`*.TEXT:10`	C250		Characters 2251-2500 of the document text	For DOKTYP<>TEXT, the parameter must be empty.
`*.PRAKTIV`	"0" or "1"	M/I	Show document during inspection?	If the value "1" is transferred to this parameter, the document is transmitted to the inspection requirement or the corresponding inspection step characteristic when the inspection step is created and is thus available during the inspection.

2.2.14 Inspection requirements

The following dialogs are available for creating or modifying inspection requirements:

- `CPAN.MODIFY` to create or modify an inspection requirement
- `CPAN.INSERT` to create an inspection requirement
- `CPAN.UPDATE` to modify an inspection requirement
- `CPAN.DELETE` to delete an inspection requirement
- `CPAN.STORNO` to cancel an inspection requirement
- `CPAN.AKTIVIEREN` to activate an inspection requirement

The parameter `CPAN.PPS:REF` is crucial for communicating with an external system. The corresponding inspection requirement is identified with this reference number.
To illustrate the user fields completed via the interface in HYDRA, the user fields must have been configured accordingly (object = "CPAN"; user field key = data type (CPAN.RECTYP).

WARNING!! **HYDRA-CAQ provides the option to copy the user fields of the inspection plan to the inspection requirement when you create an inspection requirement. This functionality is controlled in an option (option 1058). If this functionality is activated, you must guarantee that the numbers of the user fields of inspection plan and inspection requirement are identical.**

Example: The interface populates the user fields CPPL.FU:1 to CPPL:FU:14 of the inspection plan header. User fields CPAN.FU:1 to CPAN.FU:14 are passed to the inspection requirements. As a result, user fields 1 to 14 are shown in an inspection requirement based on an inspection plan.
In this case, when configuring the user fields, make sure that the field types for the user fields 1 to 14 in both the inspection plans and in the inspection requirements match.

2.2.14.1 Method calls with the PPS reference number (default)

For the `CPAN.MODIFY` method, the system first checks whether a non-completed inspection requirement with the PPS reference number (`CPAN.PPS:REF`) already exists for the data type (`CPAN.RECTYP`) in the area concerned (`CPAN.BER`). In this case, the inspection requirement that was found is modified according to the parameters transferred.
If the search for a non-completed inspection requirement was not successful, a new inspection requirement is created.
If you would like to make sure that there is exactly only one inspection requirement with the PPS reference number in the system, then you should call up the method `CPAN.AKTIVIEREN` before calling `CPAN.MODIFY`. Doing so will first "reactivate" an existing but already completed inspection requirement.
For the `CPAN.INSERT`, `CPAN.UPDATE`, `CPAN.DELETE`, `CPAN.STORNO` and `CPAN.AKTIVIEREN` methods, the program searches for an inspection requirement with the transferred reference number (parameter `CPAN.PPS:REF`) for the data type (`CPAN.RECTYP`) in the relevant area (parameter `CPAN.BER`) that has not yet been completed. If none are found, the search is repeated, however the status of the inspection requirement is now ignored.
If the search returns several inspection requirements, in each case the inspection requirement created last will be used. Each method is then applied to the data record that was found.

2.2.14.2 Method calls without a PPS reference number

It is also possible to create inspection requirements without a PPS reference number. However, when using this call type, it becomes difficult later on to identify orders that have already been created (for example when calling the methods `CPAN.UPDATE`, `CPAN.DELETE`, `CPAN.STORNO` or `CPAN.AKTIVIEREN`).

When the `CPAN.MODIFY` method is called, the program searches for an uncompleted inspection requirement in the corresponding area (`CPAN.BER`) and matching the current data type (`CPAN.RECTYP`), in accordance with the parameters configured in system option 3. If none are found, a new inspection requirement is created.

For the method calls `CPAN.DELETE`, `CPAN.STORNO` or `CPAN.AKTIVIEREN` without the PPS reference number, the data type (`CPAN.RECTYP`), the area (`CPAN.BER`) and the internal HYDRA inspection requirement number (`CPAN.PANNR`) must be transferred. An inspection requirement can be clearly identified using these parameters. These parameters can also be used for the method `CPAN.UPDATE`.
The parameter `CPAN.PANNR` is an internal HYDRA key field. The system assigns this key field automatically when a new inspection requirement is generated. If the system is customized accordingly there is the option to identify the parameter `CPAN.PANNR` using alternative key fields. The training course CUT-IMI provides the required basics and skills.

2.2.14.3 Order number

You use the CAQ system option "1115" to define if the ADE-CAQ integration mode is activated.

The mode is activated by default. This means: the field *Order number* of the inspection requirement must absolutely include a value. To guarantee this, the order number is automatically generated when the number is not transferred via interface or when an inspection requirement is manually created without order number. The system then automatically adds a "Q" in front of the order number. The other n characters are derived from the last n places of the PPS reference number. In this case, the entry of the PPS reference number is mandatory and the number must at least have the required order number length. If the PPS reference number is too short, the system adds leading zeros before the Q. For example with an order number of 8 digits: 0000000Q.

If the integration mode is not activated, no order number is generated. The field *Order number* then remains empty. This is possible because the system does not generate an order for the inspection requirement with this mode.

2.2.14.4 Field definitions

Dialog: `CPAN.*`				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
`*.RECTYP`	"FEP", "WEP", "WAP", "EMU"	M/I/U/D/S/A	Data type of inspection requirement	FEP = Production order WEP = Goods receipt WAP = Goods issue EMU = Initial sample inspection

`*.BER`	C10C10	M/I/U/D/S/A	Area for which the inspection requirement applies.	An area with the corresponding area ID must exist in HYDRA. By default, these are: E = Goods receipt F = Production A = Goods issue EMU = Initial sample
`*.PANNR`	N9	(M/I/U/D/S/A)*	Internal HYDRA inspection requirement number	
`*.PPS:REF`	C250	(M/I/U/D/S/A)*	PPS reference number	
`*.PPS:ZCHR1`	C250		PPS addition	
`*.ANR`	C250	(M/I/U/D/S/A)*	Order number	The database permits an entry of up to 250 characters in this field. But this field is bound to the *Length of order number* specified in the HYDRA basic settings and you must therefore respect the length specified there. Do not use any special characters.
`*.ATK`	C50	(M/I/U/D/S/A)*	Article number	An article with the corresponding combination of article number and drawing issue number must exist in HYDRA.
`*.ATKIDX`	C50	(M/I/U/D/S/A)*	Drawing issue number of the article	An article with the corresponding combination of article number and drawing issue number must exist in HYDRA. If required, the drawing issue number may be left empty.
`*.AGNR`	C50	(M/I/U/D/S/A)*	Operation number	
`*.AGBEZ`	C250		Operation designation/name	

`*.PPLID`	C50		Inspection plan number	Inspection plan number used to create the inspection requirement. If the inspection plan number is not specified, HYDRA automatically identifies the inspection plan using the defined criteria.
`*.PPLIDX`	C50		Inspection plan version	Inspection plan version used to create the inspection requirement. If the inspection plan number is not specified, HYDRA automatically identifies the inspection plan using the defined criteria.
`*.KDNR`	C50	(M/I/U/D/S/A)*	Customer number	For RECTYP=WEP, the parameter must be empty. A customer with the respective customer number must exist in HYDRA.
`*.HERSTNR`	C50	(M/I/U/D/S/A)*	Manufacturer number	For RECTYP<>WEP, the parameter must be empty. A manufacturer with the respective manufacturer number must exist in HYDRA.
`*.LIEFNR`	C50	(M/I/U/D/S/A)*	Supplier number	A supplier with the respective supplier number must exist in HYDRA.
`*.CNR`	C250	(M/I/U/D/S/A)*	Batch number	
`*.MNR`	C50	(M/I/U/D/S/A)*	Machine number	
`*.PANDAT`	Date		Date of the inspection requirement	If no value was transferred and a new inspection requirement is created, the field is filled with the current system date during processing.

*.PANZEI	Time		Time of the inspection requirement	If no value was transferred and a new inspection requirement is created, the field is filled with the current system time during processing.
*.PANVON	C50		Person or process that has set the inspection requirement	If no value was transferred and a new inspection requirement is created, then the user ID of the interface is entered in the field.
*.LIEFDAT	Date		Delivery date (actual)	
*.LIEFDAT:SOLL	Date		Delivery date (target)	
*.CMENGE	N12.4		Delivery quantity (actual)	
*.CMENGE:SOLL	N12.4		Delivery quantity (target)	
*.BESTNR	C250	(M/I/ U/D/ S/A)*	Purchase order number	
*.HERSTDAT	Date		Manufacturing date	
.FU:1 to.FU:5	C50		Direct user fields	
.FU:6 to.FU:10	N9		Direct user fields	
*.FU:11; *.FU:12	N12.9		Direct user fields	
*.FU:13; *.FU:14	Date		Direct user fields	

* please also see *2.2.14.1 Method calls with the PPS* reference number (default) and *2.2.14.2 Method calls without* a PPS reference number

2.2.14.5 AIP Specifics

Please note the following when using the AIP terminal.

If the AIP is used, inspection requirements are usually generated automatically for the production area when an operation is logged on.

2.2.15 Inspection points

WARNING! If you use inspection points in an inspection step, then you must collect the data for the inspection points. By default, this is only true for in-production inspections (data type RECTYP=FEP).
If customized accordingly, this function can also be made available for other data types.

The following dialogs are available for updating inspection points:

- `CPANUMP.MODIFY` to create or change inspection points
- `CPANUMP.INSERT` to create inspection points
- `CPANUMP.UPDATE` to change inspection points
- `CPANUMP.ABSCHLIESSEN` to complete inspection points
- `CPANUMP.FREIGEBEN` to release inspection points
- `CPANUMP.DELETE` to delete inspection points

The unique key for inspection points is made up of the fields `CPANUMP.RECTYP`, `CPANUMP.BER`, `CPANUMP.PANNR`, `CPANUMP.PAUNR`, `CPANUMP.EINTTYP` and `CPANUMP.EINTNR`.

There is also an alternative method available to reference the corresponding inspection requirement. Therefore, in some cases there is no need to use parameter `CPANUMP.PANNR` (see *2.2.15.1 Calculating the inspection* step reference).

Inspection points can only be created, changed and deleted for inspection steps with any status except for "completed", "cancelled" or "no characteristic".
The corresponding inspection requirement must not have the status "completed", "canceled", "no inspection step" or "no characteristic" in order to create new inspection points or to change or delete existing ones.

Please note that the field lengths indicated here are maximum values. Corresponding field lengths might be restricted for the AIP terminal.

2.2.15.1 Calculating the inspection step reference

If the inspection requirement number is unknown, the parameter `CPANUMP.PANNR` can be omitted. In that case the inspection requirement number is calculated based on the parameters `CPANUMP.RECTYP`, `CPANUMP.BER`, `CPANUMP.PPS:REF`. However, this requires that only exactly one inspection requirement exists with the PPS reference number transferred as parameter `CPANUMP.PPS:REF`.

All other requirements for referencing the inspection requirement and/or inspection step can only be implemented if the system is customized. The training course CUT-IMI provides the required basics and skills.

2.2.15.2 Field definition

Dialog: CPANUMP.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
*.RECTYP	"FEP“	I/M/ U/D/ A/F	Data type of inspection requirement	By default, only in-production inspections • FEP = production order are configured for data collection relating to inspection points. If required, other areas can also be configured accordingly. In this case, the corresponding entries (see inspection requirements) must be used.
*.BER	C10	I/M/ U/D/ A/F	Area for which the inspection requirement applies.	An area with the corresponding area ID must exist in HYDRA.
*.PANNR	N9	(I/M/ U/D/ A/F) *	Internal HYDRA inspection requirement number	In HYDRA, an inspection requirement with the respective number must exist for the type and area.

*.PAUNR	N9	(I)/(M)/ U/D/ A/F	Internal HYDRA inspection step number	In HYDRA, an inspection step with the respective number must exist for the type and area. This parameter must remain empty, if an inspection point is generated as sample of a sample group.
*.EINTTYP	"PPUNKT“	I/M/ U/D/ A/F	Type of entry	
*.EINTNR	C 5	(M)/ U/D/ A/F	Inspection point number	5-digit numeric number of the inspection point including leading zeros (example: "00014"). This parameter must remain empty if a new inspection point is to be created. As the inspection point number must always be assigned by HYDRA.
*.PPKT:EQUIP	C 20		Equipment	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field. For cavity-related inspection results recording, this field includes the tool. A tool with this number must exist in HYDRA.
*.PPKT:TPLATZ	C 20		Functional location	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field.
*.PPKT:TLOS	C50		Partial batch	It might be necessary to treat this parameter like a mandatory field, if the inspection step requires it.
*.PPKT:CNRs	C50		Batch number	It might be necessary to treat this parameter like a mandatory field, if the inspection step requires it.

*.PPKT:USERC1	C50		Inspection point user field C1	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field.
*.PPKT:USERC2	C50		Inspection point user field C2	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field.
*.PPKT:USERN1	N9		Inspection point user field N1	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field.
*.PPKT:USERN2	N9		Inspection point user field N2	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field.
*.PPKT:USERD1	Date		Inspection point user field D1	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field.
*.PPKT:USERT1	Time		Inspection point user field T1	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field.
*.PPKT:PRBGRP	C50		Sample group	If this field is completed, the inspection point represents a sample of the defined sample group. An inspection step including inspection characteristics of this sample group must exist in the inspection requirement. In this case, the parameter CPANUMP.PAUNR must not be indicated as HYDRA searches autonomously for the corresponding inspection step.

*.PPKT:PROBE	C 20		Physical sample	The corresponding inspection step defines if this field should be displayed in the input dialog and if it is an optional or mandatory field. This parameter can remain empty if a new sample is to be created for a specific sample group. Then HYDRA generates the sampe number using the number range "QMPRBNR".
*.PPKT:MNR	C 20		Target workplace	Usually, the AIP only shows inspection points of machines matching this parameter. If this parameter is empty, the AIP terminal shows all inspection points (of all machines). A workplace with the respective number must exist in HYDRA.
*.PPKT:PRODMNR	C 20		Machine producing the items for the inspection point.	A workplace with the respective number must exist in HYDRA.
*.PPKT:CMENGE	N12.4		Quantity	
*.PPKT:EGRAUS	N12.4		Scrap quantity	
*.PPKT:EGRNACH	N12.4		Rework quantity	
*.PPKT:BEM	C10		Comment	

*.PPKT:ANLURS	C50		Cause for creation	The cause for creation identifies the event triggering the generation of the inspection point. By default, HYDRA uses the following causes: • FREI - manually generated • MENGE - production quantity • ZEIT - production time • A_AN - OP logon • M_MST - machine status change • PROBE - sampling • ALW - output batch change • SW - shift change We recommend using separate causes for separate applications. To do so, a CAQ status of the type "PPKTANLURS“ must be created.
*.PPKT:ANLURSDAT	Date		Date of the event triggering the generation of inspection points	
*.PPKT:ANLURSZEI	Time		Time of the event triggering the generation of inspection points	
*.PPKT:ANLDAT	Date		Date of generation	
*.PPKT:ANLZEI	Time		Time of generation	
*.PPKT:ANLVON	C50		Person generating the inspection point	
*.PPKT:ABSDAT	Date		Date of completion	
*.PPKT:ABSZEI	Time		Time of completion	
*.PPKT:ABSVON	C50		Person completing the inspection point	

*.CNR	C 20		HYDRA batch number	If this parameter is completed, this inspection point is assigned to the referenced batch. A batch with this number must exist in HYDRA.
*.PPKT:VEKATART	C10		Catalog type of the usage decision	This parameter may only be transferred with CPANUMP.ABSCHLIESSEN. Usually, this parameter is assigned to the value "QM_PP_BEW". A catalog entry matching the catalog type, site/plant, selected set, code group and code must exist in HYDRA.
*.PPKT:VEWERK	C 4		Site/plant of the usage decision	This parameter may only be transferred with CPANUMP.ABSCHLIESSEN. Usually, this parameter is assigned to the value "0001". A catalog entry matching the catalog type, site/plant, selected set, code group and code must exist in HYDRA.
*.PPKT:VEAUSWMEN	C10		Selected set of the usage decision	This parameter may only be transferred with CPANUMP.ABSCHLIESSEN. Usually, this parameter is assigned to the value "PPKT_VE". A catalog entry matching the catalog type, site/plant, selected set, code group and code must exist in HYDRA.
*.PPKT:VECODGR	C10		Code group of the usage decision	This parameter may only be transferred with CPANUMP.ABSCHLIESSEN. Usually, this parameter is assigned to the value "01". A catalog entry matching the catalog type, site/plant, selected set, code group and code must exist in HYDRA.

`*.PPKT:VECODE`	C10		Code of the usage decision	This parameter may only be transferred with CPANUMP.ABSCHLIESSEN. Usually, this parameter is assigned to the value "A" (accepted) for a "pass" usage decision or "R" (reject) for a "failed" usage decision. A catalog entry matching the catalog type, site/plant, selected set, code group and code must exist in HYDRA.
`*.FU:1` to `*.FU:5`	C50		Direct user fields	
`*.FU:6` to `*.FU:10`	N9		Direct user fields	
`*.FU:11`; `*.FU:12`	N12.9		Direct user fields	
`*.FU:13`; `*.FU:14`	Date		Direct user fields	
`*.MOD:NOSTATUSERR`	"0" or "1"	F	The status of inspection requirements or inspection steps is not checked.	If this parameter is "1", the system allows creation and editing of inspection points, although the inspection requirement status or inspection step status is invalid (canceled, competed, etc.). If the status is invalid and the parameter is "0", an error message is issued.

* please also see *2.2.15.1 Calculating the inspection* step reference

2.2.16 Measured values/inspection results

The following dialogs are available for updating measured values:

- `CPAUMW.INSERT` to create measured values
- `CPAUMW.UPDATE` to modify measured values
- `CPAUMW.MODIFY` to modify a measured value (if the referenced measured value does not exist, it is created).

- CPAUMW.DELETE to delete measured values

The unique key for the measured values is made up of the fields CPAUMW.RECTYP, CPAUMW.BER, CPAUMW.PANNR, CPAUMW.PAUNR, CPAUMW.AFO, CPAUMW.STPRNR and CPAUMW.WERTNR.

There are also alternate methods available to reference the corresponding inspection step. With them, in some cases there is no need to use the parameters CPAUMW.PANNR or CPAUMW.PAUNR (see *2.2.16.1 Calculating the inspection* step reference).

When creating new measured values, using the parameters CPAUMW.DEVICE:TYP, CPAUMW.DEVICE:ID and CPAUMW.DEVICE:STPRNR device-specific sample number ranges can be used (see *2.2.16.2 Device-specific number ranges for* samples). In this case, parameter CPAUMW.STPRNR must not be used.

Furthermore, when a new measured value is created, parameter CPAUMW.WERTNR and also parameter CPAUMW.STPRNR must not be used (see *2.2.16.3 Dynamic calculation of sample* numbers and *2.2.16.4 Dynamic identification of the* measured value number).

Before creating or modifying measured values, you must make sure that the corresponding inspection step characteristic already exists.

Creating or modifying values for calculated characteristics is only allowed if the characteristics have a corresponding formula type (see documentation).

If a measured value violates the plausibility limits of a characteristic, it is usually rejected (see also parameter CPAUMW.MOD:NOPLAUSI).

Creating valid measured values and modifying or deleting existing measured values is only possible for inspection steps with any status except for "completed", "canceled" or "no characteristic" (please also see parameter CPAUMW.MOD:NOSTATUSERR).

Please note that the field lengths indicated here are maximum values. Corresponding field lengths might be restricted for the AIP terminal.

2.2.16.1 Calculating the inspection step reference

If the inspection requirement number is unknown, the parameter CPAUMW.PANNR can be omitted. The inspection requirement number is in that case calculated based on the parameters CPAUMW.RECTYP, CPAUMW.BER und CPAUMW.PAUNR.

If the inspection step number is unknown, the parameter `CPAUMW.PAUNR` can be omitted. The inspection step is then calculated based on the parameters `CPAUMW.RECTYP`, `CPAUMW.BER`, `CPAUMW.PANNR` and `CPAUMW.AFO`.

If the parameter `CPAUMW.PANNR` and the parameter `CPAUMW.PAUNR` are unknown, then they can also be calculated based on the parameters `CPAUMW.RECTYP`, `CPAUMW.BER`, `CPAUMW.PPS:REF` and `CPAUMW.AFO`. However, this requires that only exactly one inspection requirement exists with the PPS reference number transferred as parameter `CPAUMW.PPS:REF`.

All other requirements to reference the inspection step can only be implemented if the system is customized. The training course CUT-IMI provides the required basics and skills.

2.2.16.2 Device-specific number ranges for samples

When working with measured values, optionally you can also specify from which device they originate. A separate number range for samples can exist for each device. This may deviate from the HYDRA sample number range.

Example:

At device A, three samples and at device B two samples of an inspection step characteristic were recorded with the following reference:

```
CPAUMW.RECTYP=FEP
CPAUMW.BER=F
CPAUMW.PANNR=5
CPAUMW.PAUNR=7
CPAUMW.AFO=30
```

In this case, the entries of the HYDRA-CAQ sample table can appear as follows:

RECTYP	BER	PANNR	PAUNR	OP se-quence	STPRNR	DEVICE:ID	DEVICE:STPRNR
FEP	F	5	7	30	1	A	1
FEP	F	5	7	30	2	B	1
FEP	F	5	7	30	3	B	2
FEP	F	5	7	30	4	A	2

FEP	F	5	7	30	5	A	3

If device-specific sample number ranges are used, the sample can be referenced using the parameters `CPAUMW.DEVICE:TYP`, `CPAUMW.DEVICE:ID` and `CPAUMW.DEVICE:STPRNR`. In this case, parameter `CPAUMW.STPRNR` must not be used.

WARNING! Device-specific number ranges for samples can only be used if inspection steps are recorded in relation to samples instead of inspection points.
The sample number is largely specified by the inspection point if inspection steps are recorded in relation to inspection points (exception: cavity-related data collection for samples).

2.2.16.3 Dynamic calculation of sample numbers

If the parameter `CPAUMW.STPRNR` (or the parameters `CPAUMW.DEVICE:TYP`, `CPAUMW.DEVICE:ID` and `CPAUMW.DEVICE:STPRNR`) is omitted as well, a check is run to determine whether the last sample of the referenced characteristic was completed (manually or by reaching the sample size). In this case, a new sample is generated for the measured value. Otherwise, the measured value is added to the last sample.

WARNING! Sample numbers can only be identified dynamically if inspection steps are recorded in relation to samples instead of inspection points.
The sample must always be referenced directly for inspection steps recorded in relation to inspection points.

To modify and delete measured values, the sample number (parameters `CPAUMW.STPRNR` or `CPAUMW.DEVICE:TYP`, `CPAUMW.DEVICE:ID` and `CPAUMW.DEVICE:STPRNR`) must always be specified.

All other requirements to reference the sample can only be implemented by customizing the system. The training course CUT-IMI provides the required basics and skills.

2.2.16.4 Dynamic identification of the measured value number

New measured values can be created even without specifying the parameter `CPAUMW.WERTNR`. In this case, measured values are simply added to the end of the specified sample (referenced by the parameter `CPAUMW.STPRNR` or the parameters `CPAUMW.DEVICE:TYP`, `CPAUMW.DEVICE:ID` and `CPAUMW.DEVICE:STPRNR`).

When modifying and deleting measured values, the measured value number (the parameter `CPAUMW.WERTNR`) must always be indicated.

All other requirements to reference the measured value can only be implemented by customizing the system. The training course CUT-IMI provides the required basics and skills.

2.2.16.5 Field definition

Dialog: `CPAUMW.*`				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
`*.RECTYP`	„FEP“, „WEP“, „WAP“, „EMU“, „PMV“	I/M/U/D	Data type of inspection requirement	FEP = Production order WEP = Goods receipt WAP = Goods issue EMU = Initial sample inspection PMV = Test equipment (gage) management
`*.BER`	C10	I/M/U/D	Area for which the inspection requirement applies.	An area with the corresponding area ID must exist in HYDRA.
`*.PANNR`	N9	(I/M/U/D) *	Internal HYDRA inspection requirement number	In HYDRA, an inspection requirement with the respective number must exist for the type and area.
`*.PAUNR`	N9	(I/M/U/D) *	Internal HYDRA inspection step number	In HYDRA, an inspection step with the respective number must exist for the type and area.
`*.AFO`	N9	I/M/U/D	Work sequence (OP sequence) number of the inspection step characteristic	In HYDRA, a characteristic with this work sequence (OP sequence) number must exist for the referenced inspection step
`*.STPRNR`	N9	(I/M/U/D) *	Internal HYDRA sample number	The sample number will be issued consecutively within an inspection step characteristic, starting at 1.

*.WERTNR	N9	(I/M/U/D) *	Internal HYDRA number of the measured value within a sample	The number of the measured value will be issued consecutively within a sample of an inspection step characteristic, starting at 1.
*.DEVICE:TYP	C50	(I/M/U/D) *	Device type at which the measured value was recorded	
*.DEVICE:ID	C50	(I/M/U/D) *	Device's identifier at which the measured value was recorded	This parameter can only be used in combination with the parameter DEVICE:TYP.
*.DEVICE:STPRNR	N9	(I/M/U/D) *	Sample number at the device at which the measured value was recorded	This parameter can only be used in combination with the parameters DEVICE:TYP and DEVICE:ID.
*.MOD:AKTNUM	„1“	(I/M/U/D)	Flag indicating if the corresponding inspection point must be updated.	**This parameter may and must only be transferred if the corresponding characteristic is collected based on inspection points.**
*.NUM:EINTTYP	"PPUNKT“	(I/M/U/D)	Type of inspection point that must be updated.	**This parameter may and must only be transferred if the corresponding characteristic is collected based on inspection points.**
*.NUM:EINTNR	C 5	(I/M/U/D)	Number of the inspection point that must be updated.	**This parameter may and must only be transferred if the corresponding characteristic is collected based on inspection points.** This parameter includes the number of the corresponding inspection point (5-digit, including leading zeros).
*.MOD:AKTNPSTA	„1“	(I/M/U/D)	Flag indicating if the status of the corresponding inspection point must be updated.	**This parameter may and must only be transferred if the corresponding characteristic is collected based on inspection points.**

`*.UNGUELTIG`	"0" or "1"	I/M/U	Flag showing the validity of the measured value	If this parameter is "1", the corresponding measured value is marked as invalid. It is not used for the evaluation of the corresponding characteristic or in the statistics calculation. If this parameter is "0", the corresponding measured value is considered to be valid.
`*.BEM`	C250		Comment	
`*.MW`	N12.8		Measured value	The parameter must be empty for attributive characteristics.
`*.BFARG:1` to `*.BFARG:10`	N12.8		Arguments for calculated characteristics	These parameters are only available from CAQ 8.2 onwards. These parameters may only be used for enhanced calculated characteristics. Only arguments that are part of the formula are supported. Then these parameters must be treated like a mandatory field.
`*.STPRUMF`	N9		Sample size	The parameter must be empty with variable characteristics.This parameter specification is optional. If it is empty, the sample size of the corresponding characteristic is assumed.
`*.ERRMENGE`	N9		Number of non-conforming units	The parameter must be empty with variable characteristics.
`*.BEWKATART:1`	C10		Catalog type of the assessment catalog	This parameter is only available as of CAQ 8.2. This parameter may only be used for characteristics evaluated based on catalogs. The entry for the corresponding characteristic referenced in the HYDRA assessment catalog must be available. Usually, the value of this parameter is assigned to "BEW_CODE_1".

*.BEWWERK:1	C 4		Site/plant of the assessment catalog	This parameter is only available as of CAQ 8.2. This parameter may only be used for characteristics evaluated based on catalogs. The entry for the corresponding characteristic referenced in the HYDRA assessment catalog must be available. Usually, the value of this parameter is empty.
*.BEWAUSWMEN:1	C10		Selected set of the assessment catalog	This parameter is only available as of CAQ 8.2. This parameter may only be used for characteristics evaluated based on catalogs. The entry for the corresponding characteristic referenced in the HYDRA assessment catalog must be available. Usually, the ID of the selected set matches the ID of the parameter "selected set" of the characteristic.
*.BEWCODGR:1	C10		Failure group of the assessment catalog	This parameter is only available as of CAQ 8.2. This parameter may only be used for characteristics evaluated based on catalogs. The entry for the corresponding characteristic referenced in the HYDRA assessment catalog must be available.
*.BEWCODE:1	C10		Defect/failure code of the assessment catalog	This parameter is only available as of CAQ 8.2. This parameter may only be used for characteristics evaluated based on catalogs. The entry for the corresponding characteristic referenced in the HYDRA assessment catalog must be available.

`*.MNR`	C50		Machine that the measured value belongs to	This parameter specification is optional. The machine number is managed at sample level. By specifying this parameter, the machine is assigned to the current sample. A machine with the corresponding machine number must exist in HYDRA.
`*.PNR`	C50		User ID for the inspector	
`*.KNR`	C50		The inspector's badge number	Im HYDRA muss eine Person mit der entsprechenden Kartennummer existieren.
`*.MWDAT`	Date		Date when the measured value was recorded	If this parameter is empty, the current system date is entered.
`*.MWZEI`	Time		Time when the measured value was recorded	If this parameter is empty, the current system time is entered.
`*.MWID`			Measured value ID	Usually, this parameter is only used if inspection results are imported via MDI. In exceptional cases, it can also be used for the identification of the measured value. This parameter must be unique throughout the system.
`*.MOD:NOPLAUSI`	"0" or "1"	F	No validation check	If this parameter is "1", the measured value is not checked. No inspection will be conducted against the plausibility/validation limits defined in the inspection characteristic.

*.MOD:NOSTATUSERR	"0" or "1"	F	The status of inspection requirements or inspection steps is not checked.	If this parameter is "1", the system will allow a new measured value to be created for an invalid status for inspection requirements or inspection steps (cancellation, finished, etc.). However, in this case the measured value will be marked as invalid, irrespective of the content of the parameter CPAUMW.UNGUELTIG. If the status is invalid and the parameter is "0", an error message is issued. Modifying or deleting existing measured values is generally not possible for an invalid status.

* in this regard see chapter *2.2.16.1 Calculating the inspection* step reference
and *2.2.16.2 Device-specific number ranges for* samples
and *2.2.16.3 Dynamic calculation of sample* numbers
and *2.2.16.4 Dynamic identification of the* measured value number

2.2.17 Complaint header data

The following dialogs are available for updating complaint header data:

- CREKAUFT.INSERT   to create entries for the complaint header data
- CREKAUFT.UPDATE   to change entries for the complaint header data
- CREKAUFT.DELETE   to delete entries for the complaint header data

The unique key for the complaint header data is made up of the fields CREKAUFT.BER and CREKAUFT.REKAUFT.

Dialog: CREKAUFT.*				
Parameter	**Type**	**Mandatory**	**Contents**	**Description**
*.RECTYP	"REK"	I/U/D	Fixed identification for complaint data	
*.BER	C10	I/U/D	Area for which the complaint applies.	An area with the corresponding area ID must exist in HYDRA.

*.REKAUFT	C50	(I/U)/D	Complaint order number	This parameter and the area clearly identify the complaint. This number can be generated by the system if HYDRA was configured accordingly. Therefore, the number does not need to be transferred. WARNING! Later, it may be necessary to identify the complaint details using this number.
*.REKART	C50	F	Complaint type	Only identifiers defined in HYDRA can be used. By default, these are the following: • "KUNDE" – customer complaint • "LIEFERANT" – supplier complaint • "INTERN" – internal complaint
*.REKDAT	Date		Date of the complaint	
*.REKZEI	Time		Time of the complaint	
*.REKVON	C50		Complaint created by	
*.EXTNR	C250		External complaint number	
*.STA	C50	F	Complaint status	Only identifiers defined in HYDRA can be used. By default, these are the following: • "ERFASST" – entered • "BEARBEIT" – being processed • "ABGESCHL" – finished If the status is not indicated when creating a complaint, the default status is used.

`*.BEF`	C50	F	Complaint findings	Only identifiers defined in HYDRA can be used. By default, these are the following: • "UNBESTIMMT" - uncertain • "UNGERECHTF" - not justified • "TEILWGER" - partly justified • "GERECHTF" - justified The standard findings/results are used if no findings are indicated when the complaint is created.
`*.REKLAM:TYP`	C50	F	Type of complaining par-ty.	The following values are valid: • "PERSON" • "ABTEILUNG" - department • "LIEFERANT" - supplier • "KUNDE" - customer • "HERSTELLER" - manufacturer
`*.REKLAM:NR`	C50		Identifier of the com-plaining party	The corresponding entry (depending on the type) must be defined in HYDRA.
`*.ANSPRP:TYP`	C50	F	Type of contact person.	The following values are valid: • "PERSON" • "ABTEILUNG" - department • "LIEFERANT" - supplier • "KUNDE" - customer • "HERSTELLER" - manufacturer
`*.ANSPRP:NR`	C50		Identifier of the contact person	The corresponding entry (depending on the type) must be defined in HYDRA.
`*.VERANT:TYP`	C50	F	Type of party responsib-le	The following values are valid: • "PERSON" • "ABTEILUNG" - department • "LIEFERANT" - supplier • "KUNDE" - customer • "HERSTELLER" - manufacturer
`*.VERANT:NR`	C50		Identifier of the party in charge	The corresponding entry (depending on the type) must be defined in HYDRA.
`*.ZIELDAT`	Date		Target date of the com-plaint	

*.ZIELZEI	Time		Target time/date of the complaint	
*.ISTDAT	Date		Actual date of the com-plaint	
*.ISTZEI	Time		Actual time of the com-plaint	
*.KST	C50		Cost center	
*.LAGER	C50		Warehouse	
*.LIEFDAT	Date		Delivery date	
*.LIEFNR	C50		Delivery note number	
*.PPS:REF	C250		PPS reference number	
*.ZUSCHR:1	C250		Additional field 1	
*.ZUSCHR:2	C250		Additional field 2	
*.ZUSCHR:3	C250		Additional field 3	
*.ZUSCHR:4	C250		Additional field 4	
*.ZUSCHR:5	C250		Additional field 5	

2.2.18 Complaint details

The following dialogs are available for updating complaint details:

- CREKDET.INSERT to create entries of the complaint details
- CREKDET.UPDATE to change entries of the complaint details
- CREKDET.DELETEto delete entries of the complaint details

The unique key for the complaint details is made up of the fields CREKDET.RECTYP, CREKDET.BER, CREDET.REKAUFT and CREKDET.REKDETNR.
In the process, parameter CREKDET.REKDETNR only needs to be transferred if you want to change or de-lete a complaint detail that has already been created.

Before creating or changing complaint details, you must make sure that the corresponding complaint header data already exists.

Dialog: `CREKDET.*`				
Parameter	**Type**	**Man-dato-ry**	**Contents**	**Description**
`*.RECTYP`	"REK"	I/U/D	Fixed identification for complaint data	
`*.BER`	C10	I/U/D	Area for which the com-plaint applies.	An area with the corresponding area ID must exist in HYDRA.
`*.REKAUFT`	C50	I/U/D	Complaint order number	This parameter and the area clearly identi-fy the complaint.
`*.REKDETNR`	N9	(I/U)/D	Complaint detail	Numeric value that identifies a complaint's detail. This value is automatically generated when a complaint detail is created.
`*.ATK`	C50		Article number	An article with the corresponding combina-tion of article number and drawing issue number must exist in HYDRA.
`*.ATKIDX`	C50		Drawing issue number	An article with the corresponding combina-tion of article number and drawing issue number must exist in HYDRA. If required, the drawing issue number may be left empty.
`*.LIEFNR`	C50		Supplier number	A supplier with the corresponding supplier number must exist in HYDRA.
`*.BESTNR`	C250		Purchase order number	
`*.SERIENR`	C250		Serial number	
`*.CNR`	C250		Batch number	

*.STA	C50	F	Status of the complaint detail	Only identifiers defined in HYDRA can be used. By default, these are the following: • "ERFASST" – entered • "BEARBEIT" – being processed • "ABGESCHL" – finished If the status is not indicated when creating a complaint, the default status is used.
*.BEF	C50	F	Findings of the com-plaint detail	Only identifiers defined in HYDRA can be used. By default, these are the following: • "UNBESTIMMT" - uncertain • "UNGERECHTF" - not justified • "TEILWGER" - partly justified • "GERECHTF" - justified • "GARANTIE" • "KULANZ" The standard findings/results are used if no findings are indicated when the com-plaint is created.
*.VERANT:TYP	C50	F	Type of party responsib-le	The following values are valid: • "PERSON" • "ABTEILUNG" - department • "LIEFERANT" - supplier • "KUNDE" - customer • "HERSTELLER" - manufacturer
*.VERANT:NR	C50		Identifier of the party in charge	The corresponding entry (depending on the type) must be defined in HYDRA.
*.CMENGE:LIEF	N12.4		Delivery quantity	
*.CMENGE:REKL	N12.4		Quantity subject to com-plaint	
*.CMENGE:PRUEF	N12.4		Quantity checked	
*.CMENGE:ERR	N12.4		Defective quantity	
*.CMENGE:PPM	N12.4		Parts per million	

*.WERT:LIEF	N12.2		Delivery value	
*.WERT:REKL	N12.2		Value of the goods subject to complaint	
*.REKANTEIL	N12.4		Percentage of complaints	
*.ERRANTEIL	N12.4		Percentage of defects	
*.PPL:BER	C10		Area in the corresponding inspection plan	An area with the corresponding area ID must exist in HYDRA.
*.PPL:PPLNR	N9		Number of the corresponding inspection plan	An inspection plan with the respective number must exist in HYDRA.
*.PRPAN:BER	C10		Area of the corresponding source inspection requirement	An area with the corresponding area ID must exist in HYDRA.
*.PRPAN:PANNR	N9		Number of the corresponding source inspection requirement	An inspection requirement with the respective number must exist in HYDRA.
*.WEPAN:BER	C10		Area of the corresponding goods receipt inspection requirement	An area with the corresponding area ID must exist in HYDRA.
*.WEPAN:PANNR	N9		Number of the corresponding goods receipt inspection requirement	An inspection requirement with the respective number must exist in HYDRA.
*.ZUSCHR:1	C250		Additional field 1	
*.ZUSCHR:2	C250		Additional field 2	
*.ZUSCHR:3	C250		Additional field 3	
*.ZUSCHR:4	C250		Additional field 4	
*.ZUSCHR:5	C250		Additional field 5	

3 Uploads to the external system

3.1 Description of the interface

An upload program writes the data in an interface file. This file can then be processed by an interface routine of the external system.

3.1.1 Data record structure

The ASCII file used to upload results is structured in tabular form and is made up of a header and several data rows.

```
Header
Data row 1
Data row 2
Data row 3
```

Headers and data rows may contain several cells. These data fields are separated via a vertical line ("|", ASCII 124). The data field itself may therefore not include such a sign.

```
Header1|Header2|Header3
Cell1|Cell2|Cell3
Cell1|Cell2|Cell3
Cell1|Cell2|Cell3
```

Each data row is the equivalent of one data record. Each data record must only contain as many cells as were defined in the header. The header defines how data fields are assigned to columns.

```
CPAN.RECTYP|CPAN.BER|CPAN.PANNR|CPAN.PPS:REF|CPAN.EGR:MENGE
FEP|F|123|A2345:10|23
WEP|E|126|WE34245|43.2
FEP|F|118|A2345:20|2340
```

The number of columns and their order are selected randomly and is defined in the headers.

3.1.2 Conventions used to present the various data types

Details about the positions in numeric and text fields only specifies the maximum field length. The field content itself is stored in "compressed" form, which means leading zeros or subsequent spaces are suppressed in the output.

<table>
<tr><th>Data type</th><th>Format</th><th>Examples</th></tr>
<tr><td>N<n></td><td>Numbers, a maximum of <n> positions</td><td>|2449|</td></tr>
<tr><td>N<x>.<y></td><td>Decimal number, a maximum of <x> positions before the comma. After the comma, only <y> positions are practical. A dot is the decimal separator.</td><td>|30.5000|</td></tr>
<tr><td>C<n>
Texts (characters)</td><td>Optional, the maximum length of <n> must be considered, though.</td><td>|Huber|</td></tr>
<tr><td>Date</td><td>MM/DD/YYYY (American format)</td><td>|12/31/2001|</td></tr>
<tr><td>Times or durations</td><td>Seconds since midnight or
HH:MM or
HH:MM:SS or
HH,DDD or
HH.DDD

H hours (as many places as required)
M Minutes (in groups of 60)
S Seconds
D Industrial or decimal minutes (in groups of 100)</td><td>|52200| or
|14:30| or
|14:30:00| or
|14,5| or
|14.5|</td></tr>
<tr><td>" "</td><td>Constant value</td><td></td></tr>
</table>

3.1.3 Calling the interface function for uploads

The "results" of the inspection are logged by the HYDRA system. The information is logged in a file on the server that is transferred as an upload to the external system.

Data class	Upload file
Inspection requirement results	pavrueck.asc
Results of the complaint	rekrueck.asc

Measures resulting from complaints or the complaint details	rmarueck.asc

You use an upload program to create the UTF-8 file (without BOM) that can be integrated in the HYDRA scheduler with the call.

Data class	Uploading program
Inspection requirement results	hypanrck.scr

3.1.4 Processing uploaded files

An interface program of the external system assumes the function of preparing the data structures for the transferred files so that they can be processed in batch mode or processed as online transactions.

A handshake logic must be implemented between the external system and HYDRA. This way, a secure data transfer can be realized where no data is lost and the transfer files are not overwritten.

Use the following processing method to safely process the files:

1. Rename the interface file into a new file. You do this in Windows NT from the "ren" or "rename" command and in UNIX using the "mv" command.

> **Please note:**
> When performing this step, do not use the copy command.
> As long as HYDRA is processing the file, it does **not** exist under the documented name.
> This ensures that the higher-level system can only access the file if HYDRA has not yet accessed it (secure handshake).

2. Copy the new file to the target system.
3. After the new file has been successfully transferred, it must be deleted on the HYDRA server.

3.2 Description of the data structures

3.2.1 Inspection requirement results

Below is a description of all of the cells transferred in the upload file.

Parameter	Type	Description

CPAN.RECTYP	C 20	Data type of inspection requirement
CPAN.BER	C10	Area of inspection requirement
CPAN.PANNR	N9	Internal HYDRA inspection requirement number
CPAN.STA	C50	Status of the inspection requirement • "ABG" finished • "SKL" Skip lot • "STO" canceled
CPAN.PPS:REF	C250	PPS reference number
CPAN.ANR	C250	Order number
CPAN.ATK	C50	Article number
CPAN.ATKIDX	C50	Drawing issue number of the article
CPAN.AGNR	C50	Operation number
CPAN.AGBEZ	C250	Operation designation
CPAN.KDNR	C50	Customer number
CPAN.LIEFNR	C50	Supplier number
CPAN.HERSTNR	C50	Manufacturer number
CPAN.CNR	C250	Batch number
CPAN.PANDAT	MM/DD/YYYY	Date of the inspection requirement
CPAN.PANZEI	|sssss|	Time of the inspection requirement, specified in seconds after mid-night
CPAN.PANVON	C50	Person or process that has set the inspection requirement
CPAN.LIEFDAT	MM/DD/YYYY	Delivery date (actual)
CPAN.LIEFDAT:SOLL	MM/DD/YYYY	Delivery date (target)

CPAN.CMENGE	N12.6	Delivery quantity (actual)
CPAN.CMENGE:SOLL	N12.6	Delivery quantity (target)
CPAN.BESTNR	C250	Purchase order number
CPAN.HERSTDAT	MM/DD/YYYY	Manufacturing date
CPAN.ABSDAT	MM/DD/YYYY	Date when the inspection requirement was completed For CPAN.STA=STO, this field can remain empty.
CPAN.ABSZEI	|sssss|	Time when the inspection requirement was completed, specified in seconds after midnight With CPAN.STA=STO, this field can remain empty.
CPAN.ABSVON	C50	Person or process that completed the inspection requirement For CPAN.STA=STO, this field can remain empty.
CPAN.ERGEB	C50	Result of the inspection requirement. The abbreviations for the inspection requirement results are defined in HYDRA. Default values are "IO" (pass) and "NIO" (fail). With CPAN.STA=STO, this field can remain empty.
CPAN.EGR:MENGE	N12.6	Quantity produced With CPAN.STA=STO, this field can remain empty.
CPAN.EGR:NACH	N12.6	Rework quantity With CPAN.STA=STO, this field can remain empty.
CPAN.EGR:AUS	N12.6	Scrap quantity With CPAN.STA=STO, this field can remain empty.

CPAN.VERWEND	C50	Usage decision of the inspection requirement. The abbreviations for the usage decisions are defined in HYDRA. Default values are: "FREIGABE" – release "BEDFREIGABE" – special permit "RUECKWEISEN" – reject "NACHARBEIT" - rework "SORTIEREN" - sort With CPAN.STA=STO, this field can remain empty.